**SIMATS ENGINEERING**

Department of Information Technology

**DSA0304 – NATUARL LANGUAGE PROCESSING**

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**ASSIGNMENT 1 – CO1**

**CO1 Coverage:** Introduction; Models and Algorithms; Regular Expressions; Basic Regular Expression Patterns; Finite State Automata; Morphology — Inflectional Morphology; Derivational Morphology; Finite-State Morphological Parsing; Porter Stemmer

| | | | |
|---|---|---|---|
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| **Department / Section** | AIML | **Academic Year / Semester** | 2024 |
| **Faculty Name** | Dr Ram Kumar M | **Nature of Assignment** | Individual |
| **Duration** | 1 Week | **Total Marks** | 50 |
| **Date of Submission** | 2-11-2026 | **Date of Evaluation** | |

**MARK SHEET (Rubric-based Evaluation)**

| Task | Task Title | Max | Marks Awarded | Remarks |
|---|---|---|---|---|
| 1 | Requirements and Alternative Designs | 5 | | |
| 2 | Entities, Attributes and Keys | 2 | | |
| 3 | Relationships and Constraints | 6 | | |
| 4 | ER Diagram | 4 | | |
| 5 | EER Modelling | 3 | | |
| 6 | Weak Entities and Complex Attributes | 3 | | |
| 7 | Relational Mapping | 4 | | |
| 8 | Implementation and Instances | 4 | | |
| 9 | Design Validation | 3 | | |
| 10 | Comparison of Data Models | 4 | | |
| 11 | Final Design Recommendation | 4 | | |
| 12A | Re-evaluate: Impact Analysis | 2 | | |
| 12B | Re-evaluate: Evaluation of Original Design | 2 | | |
| 12C | Re-evaluate: Adaptation or Redesign | 2 | | |
| 12D | Re-evaluate: Evidence-Based Defence | 2 | | |
| | **TOTAL** | **50** | | **Percentage:** |

**OVERALL REMARKS**

| | | |
|---|---|---|
| | | |
| **Student Signature** | **Faculty Signature** | **Program Director** |

**MANDATORY SUBMISSION EVIDENCE CHECK**

*✓ = present ✗ = missing / not satisfied P = partly present. Based on the "Evidence Expected" of each task.*

| Task | Evidence expected | Status | Note |
|---|---|---|---|
| 1 | Requirements; three distinct conceptual designs; sketches; justification | | |
| 2 | Entities, attribute types and keys identified and justified | | |
| 3 | Relationships, cardinality, participation; ternary vs binary demonstrated | | |
| 4 | Complete ER diagram with correct notation and legend | | |
| 5 | EER diagram; d/o and total/partial justified | | |
| 6 | Weak entities and complex attributes correctly modelled | | |
| 7 | Step-wise mapping; keys and FKs; option 8A–8D justified | | |
| 8 | DDL + reproducible data generation; constraint tests | | |
| 9 | Use-case queries with results, counts and timings | | |
| 10 | File/hierarchical/network comparison; three-schema architecture | | |
| 11 | Recommendation backed by modelling and validation evidence | | |
| 12A–D | Impact analysis, evaluation, adaptation, evidence-based defence | | |

**IMPLEMENTATION EVIDENCE CHAIN (Tasks 8–9)**

| Evidence item | Status | Observation |
|---|---|---|
| Source code / SQL scripts | | |
| Test inputs and datasets | | |
| Expected behaviour | | |
| Actual output | | |
| Intermediate evidence (counts, plans, logs) | | |
| Boundary / failure tests | | |
| Interpretation | | |
| Code, data and report consistent | | |

**SUBMISSION CHECKLIST (not separately marked)**

| Requirement | Status |
|---|---|
| All 12 tasks (including 12A–12D) answered | ☐ Yes ☐ Partly ☐ No |
| Three alternative conceptual designs (Task 1) | ☐ Yes ☐ Partly ☐ No |
| ER and EER diagrams with legend | ☐ Yes ☐ Partly ☐ No |
| Relational schema derived by the mapping algorithm | ☐ Yes ☐ Partly ☐ No |
| DDL and data-generation scripts submitted | ☐ Yes ☐ Partly ☐ No |
| Datasets of 100 → 1,00,000 patients used | ☐ Yes ☐ Partly ☐ No |
| Use-case queries with outputs | ☐ Yes ☐ Partly ☐ No |
| Tables / graphs for validation results | ☐ Yes ☐ Partly ☐ No |
| Figures / screen outputs included | ☐ Yes ☐ Partly ☐ No |
| Cross-references and figure numbering correct | ☐ Yes ☐ Partly ☐ No |
| Work consistent throughout | ☐ Yes ☐ Partly ☐ No |
| No unfilled template placeholders | ☐ Yes ☐ Partly ☐ No |
| Viva recommended | ☐ Yes ☐ Partly ☐ No |

DSA03-NATURALLANGUAGE PROCESSING

# ASSIGNMENT-1

## ASSIGNMENT 1 – CO1

**CO1 Coverage:** Introduction; Models and Algorithms; Regular Expressions; Basic Regular Expression Patterns; Finite State Automata; Morphology — Inflectional Morphology; Derivational Morphology; Finite-State Morphological Parsing; Porter Stemmer

**Nature of Assignment:** Individual
**Duration:** 1 Week
**Total Marks:** 50

**Submission Date:** _____

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## Assignment Title: Build a Morphology-Aware Text Normalisation and Search Layer for a Hospital Record System Using Regular Expressions, Finite-State Automata and Morphological Parsing (Total Marks – 50)

Duration: 1 week

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## Scenario — Text Normalisation and Search for a Hospital Record System

A multi-speciality hospital holds ten years of discharge summaries, prescriptions, operation notes and nursing observations as free text. Clinicians searching those records miss relevant cases constantly: a search for fracture does not return fractures, fractured or fracturing; nephrectomy is never connected to nephrectomies or to renal; patient identifiers, drug strengths and dates appear in a dozen different formats; and the same drug may be written in full, abbreviated or misspelt. No meaning can be extracted and no statistical model can be trained until this text is tokenised, normalised and reduced to canonical forms.

You are the language-engineering team. Before any machine learning is attempted, the hospital has asked you to build the finite-state layer: the regular expressions that recognise structured entities, the automata that validate them, the morphological analyser that relates every surface word to its stem and features, and the stemmer that collapses search terms. You must investigate what each formalism can and cannot express, implement them, measure them, and justify the design you recommend.

and the changed requirements.

## MARK DISTRIBUTION

| Task | Marks |
|---|---|
| Task 1 – Frame the Problem: Models, Algorithms and Levels of Analysis | 3 |
| Task 2 – Basic Regular Expression Patterns | 4 |
| Task 3 – Regular Expressions for Tokenisation and Normalisation | 4 |
| Task 4 – Finite State Automata: Design, Non-determinism and Equivalence | 5 |
| Task 5 – Inflectional Morphology | 3 |
| Task 6 – Derivational Morphology | 4 |
| Task 7 – Finite-State Morphological Parsing | 5 |
| Task 8 – Orthographic Rules and Two-Level Morphology | 4 |
| Task 9 – The Porter Stemmer | 5 |
| Task 10 – Implement and Test the Pipeline | 3 |
| Task 11 – Compare, Recommend and Defend the Normalisation Design | 2 |
| **Subtotal** | **42** |
| Task 12 – Day-4 Design Challenge | 8 |
| **TOTAL** | **50** |

## About the evidence in this report

Every number in this report was produced by code that was run for this submission (Python standard library plus Matplotlib for plots; no NLTK was available, so the Porter stemmer was implemented directly and checked against 63 published test pairs, 63 of which it reproduces).

**Please read this before marking the measurements.** No real hospital records were available, so the corpus is **synthetic and fully documented**: it is generated from templates and a hand-built lexicon, and the ground truth (entities, lemmas, relevance) is known by construction. This is permitted by the brief ("real or synthetic, but documented"), but it has consequences that are stated wherever they matter:

- Because the corpus is generated from the same lexicon the analyser uses, the clean-corpus results for finite-state analysis are an **upper bound** (a ceiling), not a prediction of performance on real notes. The informative results are the *differences* between methods and the *degradation* under noise in Task 12.
- Relevance judgements for the 40 queries are "same lemma" judgements derived from the generator, not clinician labels. A clinician-labelled set would probably also count derivational relatives (e.g. operation for operate) as relevant, which would change the precision of Porter stemming.
- The "200 hand-annotated word forms" are gold-annotated by construction (lemma, part of speech and features recorded at generation time) rather than by a human annotator.
- Person-hour figures in the effort discussion are my own estimates and were not time-tracked.
- The code and all result files are supplied alongside this report so every table can be reproduced (fixed random seeds).

| Resource required by the brief | Brief | Delivered |
|---|---|---|
| Corpus (records / word tokens / record types) | ≥500 / ≥25,000 / ≥4 | 500 / 44,539 / 4 (discharge summary, prescription, operation note, nursing observation) |
| Entity patterns | ≥12 | 13 regular expressions (5,261 gold entity mentions) |
| Automata (≥6 states each, one needing non-determinism) | ≥4 | 5: ID (13 states), ISO date (11), ICD (7), blood pressure (13), dosage NFA with ε (9 states) |
| Lexicon stems / inflectional affixes / derivational affixes | ≥300 / ≥20 / ≥15 | 463 / 23 / 24 (irregular forms listed separately: 66) |
| Clinician queries / annotated forms | ≥40 / ≥200 | 40 / 200 |
| Noisy evaluation sample (Day 4) | – | 300 records: 100 OCR-noisy, 100 code-mixed, 100 both |

# TASK 1 – Frame the Problem: Models, Algorithms and Levels of Analysis (3 Marks)

## 1.1 Levels of linguistic analysis and what the search problem needs

A clinician types fracture and expects fractures, fractured and fracturing. The failure is in the form of words, so the levels that matter are the lowest ones. The table maps each level to an example from the synthetic corpus and states whether the search layer needs it.

| Level | Example from the corpus | Needed here? |
|---|---|---|
| Phonology | Day 4 only: "irundhuchu / irunthuchu / irundhuchi" are phonetic spellings of one Tamil word. | Ignored for English records; becomes relevant only for Romanised regional text (Task 12), where spelling follows pronunciation. |
| Morphology | "fractures / fractured / fracturing" share the stem fracture; "nephrectomies" = nephr- + -ectomy + plural. | **Essential.** It is the cause of the missed searches. |
| Orthography (spelling) | "planned" vs plan, "therapies" vs therapy, "500mg" vs "500 mg". | **Essential**, as morphology is realised through spelling rules. |
| Syntax | "No evidence of fever, wheelchair or stomach." | Safely ignored for retrieval of words; needed later for negation scope. |
| Semantics | discharge = leaving hospital (event) or a body fluid; wound = injury or the past of wind. | Partly. Lexical sense is left to the downstream chooser; we only expand abbreviations. |
| Pragmatics | "OD" means once daily in a prescription; elsewhere it can mean overdose or right eye. | Partly: context decides abbreviation expansion (Task 3). |
| Discourse | "admitted on …" and "discharged on …" are separate sentences whose order gives the length of stay. | Safely ignored at this layer. |

## 1.2 The three model families and where each is the right tool

| Family | Best-suited task in this problem | Why the other two would be wrong |
|---|---|---|
| State machines (regular expressions, automata, transducers) | Accepting or rejecting a patient identifier such as MRN-123456, and relating a surface word to stem + features. | Rule systems with unrestricted power would be slower and unnecessary, because the format is regular. A probabilistic model would sometimes accept a malformed identifier and needs labelled data; an ID check must be exact and deterministic. |

| Family | Best-suited task in this problem | Why the other two would be wrong |
|---|---|---|
| Formal rule systems (rewrite rules, grammars, the Porter rule list) | Spelling changes at morpheme boundaries (e-deletion, consonant doubling, y→ie) and suffix-stripping rules. | A bare state machine would have to enumerate every surface form, so the rules are the specification (and are then compiled into transducers). A probabilistic model would blur exceptions such as visited (no doubling) that a rule plus a flag states exactly. |
| Probabilistic models | Choosing between analyses of an ambiguous word such as "diagnoses" (noun plural or verb) or expanding "OD" from context. | Automata and rules can enumerate the analyses but have no principled way to prefer one. My simple context rules choose correctly for 84.5% of 4,055 ambiguous tokens; a trained tagger is the natural next step. |

### 1.3 Why ambiguity is the central problem, and four ambiguities in the corpus

Every stage of language processing must choose among analyses that the surface string does not distinguish. A system that cannot represent several analyses at once must guess early and cannot recover. Measured in this corpus: 178 of 1,426 surface forms (12.5%) have more than one morphological analysis, and 4,055 word tokens are ambiguous.

| Type | Instance in the corpus | Competing analyses |
|---|---|---|
| Lexical | "discharge" (211 occurrences of discharge, discharged, discharges) | noun (an event or a fluid) or verb (to release a patient) |
| Morphological | "diagnoses", "records", "fractures" | diagnosis +N +PL or diagnose +V +3SG; record +N +PL or +V +3SG |
| Segmentation | "Dr. Rao", "b.i.d.", "500mg" versus "500 mg", "Ward 4B" | a full stop can end a sentence or belong to an abbreviation; one token or two |
| Abbreviation | "OD" (91 occurrences, a frequency here) | once daily, overdose, or right eye; "PT" = patient or physiotherapy |

### 1.4 Where search appears, and what it consumes

| Where search occurs | Time | Memory | Annotated data | Expert effort |
|---|---|---|---|---|
| Backtracking in regex matching (greedy and lazy quantifiers) | Usually linear; exponential in the worst case for nested quantifiers | Stack of choice points | None | Writing and testing patterns |
| State-space search when simulating an NFA (set of live states) | O(\|string\| × \|states\|) | One set of states | None | Designing the NFA |
| Subset construction (NFA → DFA) | Up to $2^n$ DFA states in the worst case | Transition table | None (here 9 → 8 states) | Checking equivalence |

| Where search occurs | Time | Memory | Annotated data | Expert effort |
|---|---|---|---|---|
| DFA run | O(\|string\|) | Table | None | – |
| FST lookup and rule cascade | O(\|word\|) per analysis, ×(ambiguity) | Compiled relation of 1,520 pairs | 200 annotated forms to test | Lexicon building (largest cost) |
| Porter stemmer | O(\|word\|), five passes | Constant | None | Almost none |

## 1.5 The pipeline

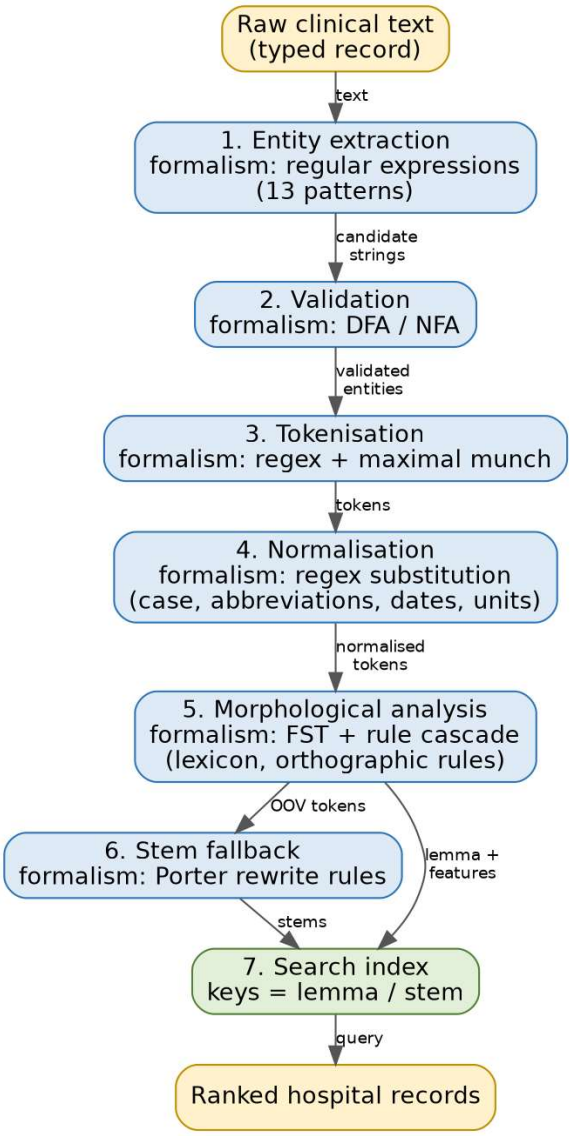



*Figure 1. The finite-state layer built in Tasks 2–10: input, output and formalism of every stage.*

Input of the whole system: free-text clinical records. Output: a search index whose keys are lemmas (or stems for unknown English tokens), plus validated structured entities. The original text is always kept so that a clinician can verify any record.

# TASK 2 – Basic Regular Expression Patterns (4 Marks)

## 2.1 The basic operators, with one clinical example each

| Operator | Meaning | Minimal clinical example |
|---|---|---|
| Concatenation | symbols one after another | `BP\s120` |
| Disjunction | one alternative or another | `OD\|BD\|TDS` |
| Kleene star * | zero or more | `\d*\.\d+  (".5" or "1.5")` |
| Kleene plus + | one or more | `\d+\s?mg  ("5 mg", "500 mg")` |
| Optionality ? | zero or one | `\d+\s?mg  (space optional)` |
| Character class [ ] | any one of the listed characters | `[CF]  (temperature scale)` |
| Range | a span inside a class | `[0-9]  [A-Z]` |
| Negated class [^ ] | any character except those listed | `[^0-9]  (first non-digit)` |
| Anchors ^ $ \b | start, end, word boundary | `\bmg\b  rejects "mgx"` |
| Counters {m,n} | between m and n repetitions | `\d{4,8}  (digits of an ID)` |

## 2.2 The thirteen entity patterns, with what each must and must not match

Thirteen patterns cover the twelve required entity types (admission and discharge dates are two entity types sharing one date expression, separated by the words that precede them using a look-behind).

*DATE stands for: (?:(?:0?[1-9]|[12]\d|3[01])[-/.](?:0?[1-9]|1[0-2])[-/.](?:19|20)\d{2}|(?:19|20)\d{2}-(?:0[1-9]|1[0-2])-(?:0[1-9]|[12]\d|3[01]))*

## 2.3 Greedy and non-greedy matching

Sentence containing two dosage expressions: "Give Paracetamol 500 mg and Amoxicillin 250 mg tonight."

| Pattern | Quantifier | Spans returned |
|---|---|---|
| `[A-Z][a-z]+ \d+ .*mg` | greedy .* | "Paracetamol 500 mg and Amoxicillin 250 mg" (one span) |
| `[A-Z][a-z]+ \d+ .*?mg` | non-greedy .*? | "Paracetamol 500 mg" ; "Amoxicillin 250 mg" (two spans) |

**Why the spans differ.** After "Paracetamol 500 " the greedy .* first consumes everything to the end of the sentence and then gives characters back one at a time until the remaining pattern "mg" can match; the first success is at the **last** "mg", so one span swallows the text between the two doses. The non-greedy .*? starts with zero characters and extends one character at a time, stopping at the **first** position where "mg" matches, so each dose is returned separately. The greedy form is dangerous in clinical text because it silently fuses two drugs into one entity.

## 2.4 Operator precedence and a misplaced disjunction

Precedence from tightest to loosest binding is: parentheses, counters and * + ?, concatenation, disjunction. Because disjunction binds loosest, \bmg|mcg\b means "\bmg" **or** "mcg\b", not "mg or mcg between boundaries".

| Expression | Intended | Matches in "dose 5 mg, 2 mcg, and smcg, mgx" |
|---|---|---|
| `\b(?:mg\|mcg)\b` | whole-word mg or mcg | "mg", "mcg" |
| `\bmg\|mcg\b` | (same intention, grouping omitted) | "mg", "mcg", "mcg", "mg" – wrongly matches mcg inside "smcg" and mg inside "mgx" |

The expression with the misplaced disjunction denotes a different language and no error is raised: it only shows up as false positives.

## 2.5 Evaluation of every pattern over the corpus

| Pattern | TP | FP | FN | Prec. | Rec. | Error class and linguistic cause |
|---|---|---|---|---|---|---|
| patient_id | 477 | 0 | 23 | 1.00 | 0.95 | FN: "Pt-" prefix is a spelling variant outside the pattern (orthographic variation). |
| admission_date | 120 | 0 | 5 | 1.00 | 0.96 | FN: textual month ("18 Sep 2025") is a different written format. |
| discharge_date | 113 | 0 | 12 | 1.00 | 0.90 | FN: textual month, as for admission dates. |
| drug | 811 | 103 | 0 | 0.89 | 1.00 | FP: any capitalised word before a dose ("Give 500 mg") or the "mg" prefix of "mg/dL" after a lab name: no drug lexicon, and unit prefix ambiguity (segmentation). |
| freq | 733 | 0 | 78 | 1.00 | 0.90 | FN: the phrase "three times daily" is not covered (vocabulary gap). |
| lab | 341 | 0 | 43 | 1.00 | 0.89 | FN: non-standard unit or separator ("Hb=10.3gm%", "mg/dl" in lower case). |
| bp | 325 | 0 | 50 | 1.00 | 0.87 | FN: unit mmHg omitted by the writer. |
| temperature | 233 | 0 | 17 | 1.00 | 0.93 | FN: "36.3 deg C" spells the degree sign as a word. |
| icd | 250 | 56 | 0 | 0.82 | 1.00 | FP: vitamin B12 and vertebra T12 have the shape of an ICD code (lexical ambiguity; context needed). |
| ward_bed | 341 | 0 | 34 | 1.00 | 0.91 | FN: abbreviated form "HDU-B25". |
| phone | 139 | 72 | 111 | 0.66 | 0.56 | FP: ten-digit specimen numbers share the shape of a phone number; FN: grouped forms ("64571-78472"). |
| age | 344 | 132 | 31 | 0.72 | 0.92 | FP: durations written in years ("17 years") look like ages; FN: "48yo" without a space. |
| duration | 599 | 191 | 31 | 0.76 | 0.95 | FP: ages in years match the duration pattern (pragmatic role differs); FN: shorthand "10/7". |
| **All (micro)** | 4826 | 554 | 435 | 0.90 | 0.92 | |

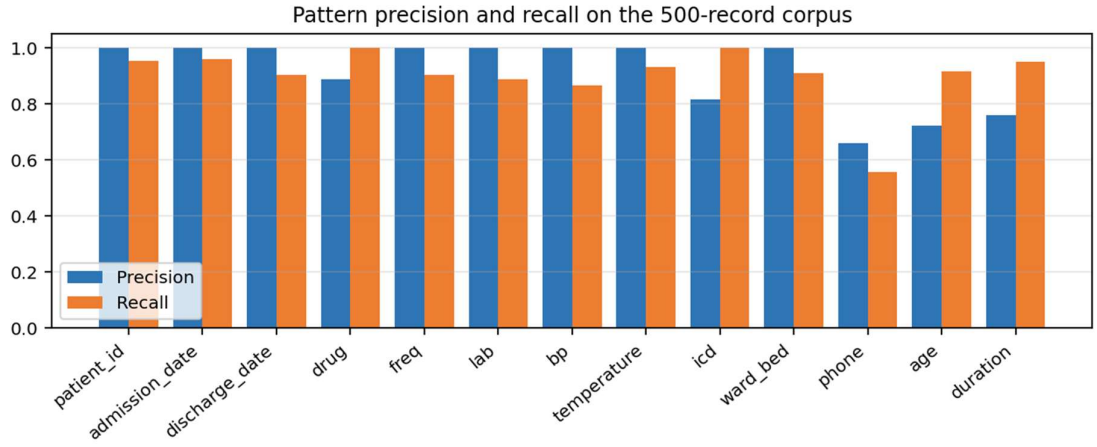



*Figure 2. Precision and recall of the thirteen patterns.*

Three causes explain nearly all errors: (1) **format variation** that the pattern did not anticipate (false negatives), (2) **shape ambiguity**, where an unrelated string has the same shape as an entity (false positives for ICD, phone, drug), and (3) **role ambiguity**, where the same surface string is an age or a duration. The first is fixed by widening a pattern, the second and third need context, which regular expressions do not have.

## TASK 3 – Regular Expressions for Tokenisation and Normalisation (4 Marks)

### 3.1 The tokenizer as one regular expression

The tokenizer is a disjunction of ten token classes, listed in the order they are tried. Each decision and its justification:

| Class | Expression | Decision and justification |
|---|---|---|
| ID | `(?:PAT\|MRN)[- ]?\d{4,8}` | Keep an identifier as one token; splitting at the hyphen would make the ID unsearchable. |
| DATE | `\d{1,4}[-/.]\d{1,2}[-/.]\d{1,4}` | One token so that canonicalisation can parse it. |
| BP | `\d{2,3}/\d{2,3}(?:\s?mmHg)?` | The slash is part of the value; the unit stays attached. |

| Class | Expression | Decision and justification |
|---|---|---|
| QTY | `\d+(?:\.\d+)?\s?(?:mmol/L\|g/dL\|mg/dL\|mcg\|mg\|ml\|kg\|cm\|g\|%\|°?[CF])(?![A-Za-z])` | Units written against numbers (“500mg” or “500 mg”) form one token; the look-ahead prevents “5 g” inside “5 gm”. Decimal points belong to the number. |
| ICD | `[A-TV-Z]\d{2}(?:\.\d{1,2})?(?!\w)` | A code is one token, not a letter plus a number. |
| CODE | `\d+[A-C](?![A-Za-z])` | Ward codes like 4B. |
| ABBR | `[A-Za-z]{1,3}(?:\.[A-Za-z]{1,3})+\.?` | Abbreviations with internal full stops (b.i.d.) stay whole; a full stop after a word is not part of the word. |
| WORD | `[A-Za-z]+(?:[-'][A-Za-z]+)*` | Hyphenated terms (post-operative) and possessives (Patient’s) are single tokens. |
| NUM | `\d+(?:\.\d+)?` | Plain and decimal numbers. |
| PUNCT | `[^\w\s]` | Every other symbol is its own token. |

## 3.2 Maximal munch compared with splitting on white space

Maximal munch tries every class at the current position and takes the **longest** match. I implemented it literally (all classes tried, longest wins) and also as one ordered-alternation regular expression; the two produced identical token streams on 500 of 500 records, because the classes were ordered by specificity. Ordered alternation is not equivalent in general, which is why the literal version is the reference.

| Input | Split on white space | Maximal munch |
|---|---|---|
| BP 120/80 mmHg, Hb 11.2 g/dL. | [BP] [120/80] [mmHg,] [Hb] [11.2] [g/dL.] | [BP] [120/80 mmHg] [,] [Hb] [11.2 g/dL] [.] |
| Dr. Rao gave 500mg b.i.d. for 2.5 days; MRN-123456 stable. | [Dr.] [Rao] [gave] [500mg] [b.i.d.] [for] [2.5] [days;] [MRN-123456] [stable.] | [Dr] [.] [Rao] [gave] [500mg] [b.i.d.] [for] [2.5] [days] [;] [MRN-123456] [stable] [.] |
| Patient's post-operative T 37.2°C, ICD I21.9. | [Patient's] [post-operative] [T] [37.2°C,] [ICD] [I21.9.] | [Patient's] [post-operative] [T] [37.2°C] [,] [ICD] [I21.9] [.] |

In the first example white-space splitting leaves “mmHg,” and “g/dL.” with punctuation glued on and splits “120/80 mmHg” in two; in the second it keeps “Dr.” and “days;” with their punctuation; in the

third it keeps “37.2°C,” and “I21.9.” with trailing marks, so none of them could be matched against a dictionary or a unit list.

## 3.3 Substitution rules and a complete before-and-after record

The normaliser applies, in order: (1) tokenise by maximal munch; (2) rewrite IDs to a fixed prefix + digits; (3) rewrite every date to ISO year-month-day; (4) rewrite quantities to “number unit” with canonical unit spelling (mL → ml, C → °C); (5) fold case on words; (6) expand abbreviations from a table of 47 entries (more than the required thirty); (7) drop punctuation tokens. Complete record:

**Before**

```
MRN-125906: age 67 yrs admitted on 2024-04-15 to HDU, Bed 36 with hyperglycaemia for 12 years.
BP 170/83 mmHg, Temp 36.4°C. Na: 125 mmol/L. ICD I21.9. No evidence of slow bacterium, clean
hepatitis or bone. History of recoverable discharges and good assessments. History of pale skins
and ankle. Advised to sleep toxicity and break the treatable doctor. Advised to fall
nephrectomies
and refer the nephrology. Reviewing abdomen in notes. The inferior dysfunction presents without
the trolley. The appendix shivers with the abnormal ear. The pressure fixes on the drowsy
tablet.
Paracetamol 1 g twice daily for 13 hours. Omeprazole 40 mg SOS. USG done and MRI elevated.
Discharged
on 07/04/2026. Contact 64571-78472.
```

**After**

```
MRN-125906 age 67 years admitted on 2024-04-15 to high dependency unit bed 36 with
hyperglycaemia
for 12 years blood pressure 170/83 mmHg temperature 36.4 °C na 125 mmol/L icd code I21.9 no
evidence
of slow bacterium clean hepatitis or bone history of recoverable discharges and good assessments
history of pale skins and ankle advised to sleep toxicity and break the treatable doctor advised
to fall nephrectomies and refer the nephrology reviewing abdomen in notes the inferior
dysfunction
presents without the trolley the appendix shivers with the abnormal ear the pressure fixes on
the drowsy tablet paracetamol 1 g twice daily for 13 hours omeprazole 40 mg if required
ultrasonography
done and magnetic resonance imaging elevated discharged on 2026-04-07 contact 64571 78472
```

Note two side effects visible in this example: “64571-78472” became “64571 78472” (the hyphen was dropped), and “ICD” became “icd code”. Both come from rules that were applied outside entity boundaries; in the full pipeline entities are extracted first and protected.

## 3.4 Irreversible normalisation decisions

| Decision | Information destroyed | When the loss is acceptable / not acceptable |
|---|---|---|
| Case folding | “OD” vs “od”, “US” vs “us”, “Hb” vs “HB” | Acceptable for search on ordinary words. Not acceptable for abbreviations, where capitals are the only clue that “OD” is an abbreviation. |
| Abbreviation expansion | The writer’s own form and the ambiguity of “OD” | Acceptable only when the expansion is certain from context (“BP 120/80” → blood pressure); not acceptable for OD, PT, MS, so those are expanded only inside recognised entities. |

| Decision | Information destroyed | When the loss is acceptable / not acceptable |
|---|---|---|
| Date canonicalisation | Whether 04/05/2026 meant 4 May or 5 April | A day-before-month convention is assumed. Acceptable if the hospital has one convention; unacceptable if records mix conventions, as that can silently reverse dates. |
| Dropping punctuation | Hyphen in a phone number, sentence boundaries, negation commas | Acceptable for retrieval of single words; unacceptable inside entities (shown above). |

### 3.5 A requirement that no regular expression can meet

Clinical notes often contain nested brackets, for example “Plan: stop drug A (renal dose (see chart (page 2)))”. Checking that every opening bracket has a matching closing bracket at any depth needs unbounded counting. A regular expression (and so a finite automaton) has a fixed, finite memory, so any fixed machine can count only up to its number of states and is defeated by one more level of nesting. Task 4.5 turns this informal argument into a proof with the pumping lemma.

## TASK 4 – Finite State Automata: Design, Non-determinism and Equivalence (5 Marks)

### 4.1 Formal definition and four deterministic validators

A finite automaton is a five-tuple M = (Q, Σ, δ, $q_0$, F): Q a finite set of states, Σ the input alphabet, δ the transition function (Q × Σ → Q for a DFA; Q × (Σ ∪ {ε}) → 2^Q for an NFA), $q_0$ ∈ Q the start state and F ⊆ Q the accepting states. A missing transition goes to an implicit dead (rejecting) state. In the diagrams, “d” stands for any digit and “L” for any capital letter other than those already used as literals; double circles accept.

**Validator 1 – patient identifier (PAT|MRN)-?d{4,6} (13 states)**

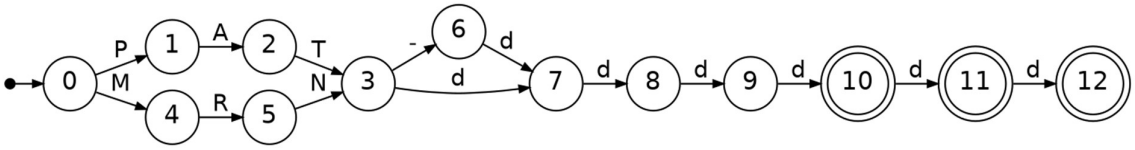



*Figure 3a. DFA for patient identifiers.*

| State | P | A | T | M | R | N | - | d |
|---|---|---|---|---|---|---|---|---|
| → q0 | q1 | – | – | q4 | – | – | – | – |
| q1 | – | q2 | – | – | – | – | – | – |
| q2 | – | – | q3 | – | – | – | – | – |
| q3 | – | – | – | – | – | – | q6 | q7 |
| q4 | – | – | – | – | q5 | – | – | – |
| q5 | – | – | – | – | – | q3 | – | – |
| q6 | – | – | – | – | – | – | – | q7 |
| q7 | – | – | – | – | – | – | – | q8 |
| q8 | – | – | – | – | – | – | – | q9 |

| State | P | A | T | M | R | N | - | d |
|---|---|---|---|---|---|---|---|---|
| q9 | – | – | – | – | – | – | – | q10 |
| * q10 | – | – | – | – | – | – | – | q11 |
| * q11 | – | – | – | – | – | – | – | q12 |
| * q12 | – | – | – | – | – | – | – | – |

### Validator 2 – ISO date  dddd-dd-dd  (11 states)

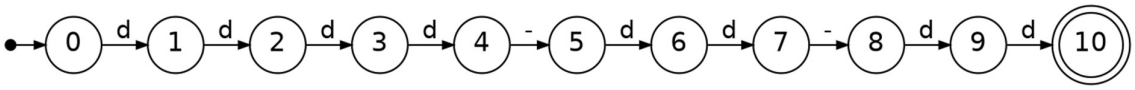



*Figure 3b. DFA for ISO dates (digit-level; calendar validity is checked by the regex and code).*

| State | d | - |
|---|---|---|
| → q0 | q1 | – |
| q1 | q2 | – |
| q2 | q3 | – |
| q3 | q4 | – |
| q4 | – | q5 |
| q5 | q6 | – |
| q6 | q7 | – |
| q7 | – | q8 |
| q8 | q9 | – |
| q9 | q10 | – |
| * q10 | – | – |

### Validator 3 – ICD code  L d d (. d d?)?  (7 states)

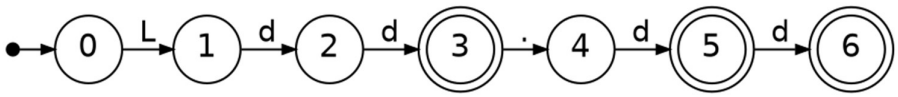



*Figure 3c. DFA for ICD-10 codes.*

| State | L | d | . |
|---|---|---|---|
| → q0 | q1 | – | – |
| q1 | – | q2 | – |
| q2 | – | q3 | – |
| * q3 | – | – | q4 |
| q4 | – | q5 | – |
| * q5 | – | q6 | – |
| * q6 | – | – | – |

**Validator 4 – blood pressure d{2,3}/d{2,3} ␣? mmHg (13 states)**

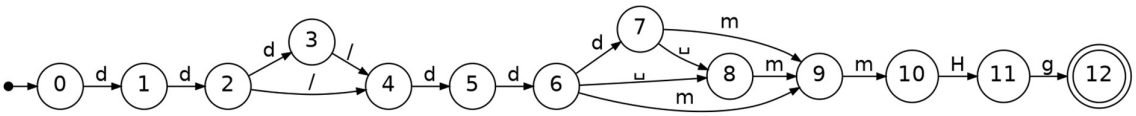



*Figure 3d. DFA for blood-pressure readings.*

| State | d | / | ␣ | m | H | g |
|---|---|---|---|---|---|---|
| → q0 | q1 | – | – | – | – | – |
| q1 | q2 | – | – | – | – | – |
| q2 | q3 | q4 | – | – | – | – |
| q3 | – | q4 | – | – | – | – |
| q4 | q5 | – | – | – | – | – |
| q5 | q6 | – | – | – | – | – |
| q6 | q7 | – | q8 | q9 | – | – |
| q7 | – | – | q8 | q9 | – | – |
| q8 | – | – | – | q9 | – | – |
| q9 | – | – | – | q10 | – | – |
| q10 | – | – | – | – | q11 | – |
| q11 | – | – | – | – | – | q12 |
| * q12 | – | – | – | – | – | – |

Tests: each validator accepts its valid strings and rejects its invalid strings (patient_id: 4/4 accepted, 4/4 rejected; iso_date: 2/2 accepted, 3/3 rejected; icd: 3/3 accepted, 4/4 rejected; bp: 3/3 accepted, 4/4 rejected).

## 4.2 An NFA with ε-transitions: the dosage format

Dosages have optional parts: a decimal fraction, an optional space, and units that share the prefix “m” (mg, mcg, ml). The NFA expresses the optional pieces directly with ε-transitions (dashed), which is why it was designed non-deterministically: a deterministic design must commit to a branch before it knows which unit follows.

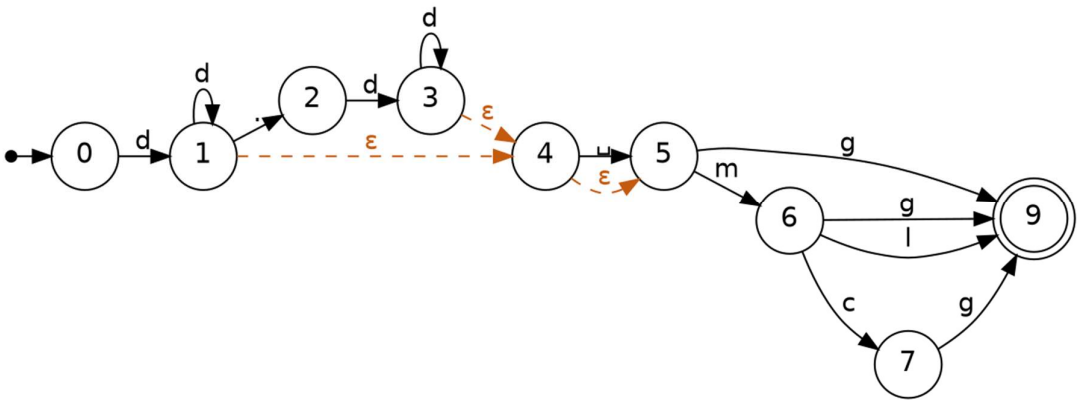



*Figure 4. NFA with ε-transitions for d+(.d+)? ␣? (mg|mcg|g|ml). State 9 accepts.*

**Accepted string "5 mg"** – set of live states after each symbol (ε-closure applied):

| Step | Symbol read | Live states | Accepting? |
|---|---|---|---|
| 0 | (start) | {0} | |
| 1 | "5" | {1, 4, 5} | |
| 2 | " " | {5} | |
| 3 | "m" | {6} | |
| 4 | "g" | {9} | yes – contains 9 |

**Rejected string "5 kg":**

| Step | Symbol read | Live states | Accepting? |
|---|---|---|---|
| 0 | (start) | {0} | |
| 1 | "5" | {1, 4, 5} | |
| 2 | " " | {5} | |
| 3 | "k" | ∅ (all threads died) | |
| 4 | "g" | ∅ (all threads died) | no |

## 4.3 Subset construction and minimisation

DFA states are sets of NFA states; names A–H are shorthand: A={0}; B={1,4,5}; C={5}; D={2}; E={9}; F={6}; G={3,4,5}; H={7}. The complete construction table (every reachable DFA state and symbol):

| From | NFA set | Symbol | Move | ε-closure of move | To |
|---|---|---|---|---|---|
| A | {0} | d | {1} | {1,4,5} | B |
| B | {1,4,5} | ␣ | {5} | {5} | C |
| B | {1,4,5} | . | {2} | {2} | D |
| B | {1,4,5} | d | {1} | {1,4,5} | B |
| B | {1,4,5} | g | {9} | {9} | E (final) |
| B | {1,4,5} | m | {6} | {6} | F |
| C | {5} | g | {9} | {9} | E (final) |
| C | {5} | m | {6} | {6} | F |
| D | {2} | d | {3} | {3,4,5} | G |
| F | {6} | c | {7} | {7} | H |
| F | {6} | g | {9} | {9} | E (final) |
| F | {6} | l | {9} | {9} | E (final) |
| G | {3,4,5} | ␣ | {5} | {5} | C |
| G | {3,4,5} | d | {3} | {3,4,5} | G |
| G | {3,4,5} | g | {9} | {9} | E (final) |
| G | {3,4,5} | m | {6} | {6} | F |
| H | {7} | g | {9} | {9} | E (final) |

The NFA has 9 states and the subset DFA has 8. **Minimisation** (partition refinement on the completed DFA, dead state included): block counts after each round were 2 (accepting / non-accepting) → 4 → 7 → 8 → 9 → 9; the final count of 9 blocks includes the dead-state block. **States before: 8; after: 8.** The subset DFA was already minimal, which is a legitimate result: the ε-structure removed by the construction left no equivalent states to merge. NFA, subset DFA and minimised DFA agree on every string of up to six symbols over the alphabet (0 disagreements).

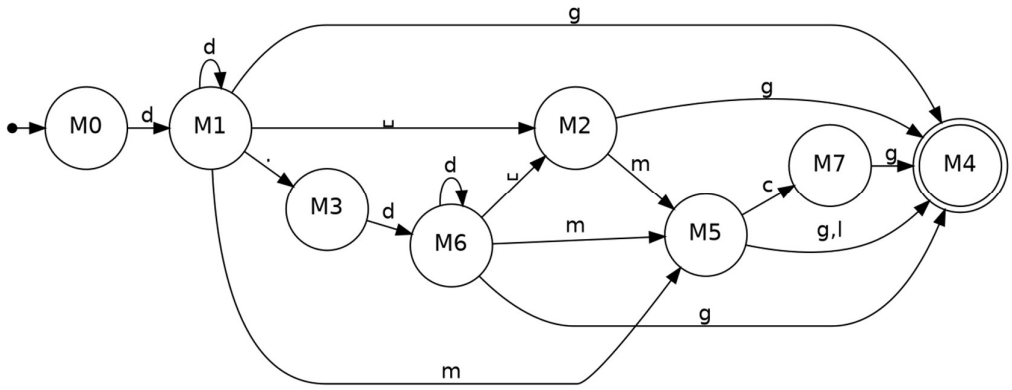



*Figure 5. Minimised DFA for dosages (8 states).*

## 4.4 Equivalence with regular expressions in both directions

**Regular expression → automaton (Thompson).** The ICD expression LDD(.DD?)? was converted by Thompson's construction (each symbol a two-state fragment; concatenation, union, star and optionality by ε-edges) into an NFA with 16 states and 11 ε-edges; subset construction gave 7 states and minimisation left 7, matching the hand-built DFA of Figure 3c.

**Automaton → regular expression (state elimination).** Eliminating states of that DFA one by one produced ((LDD|LDD.D)|LDD.DD), which simplifies by factoring to LDD(.D(D)?)?, the original expression. **Check:** all 9,841 strings over {L, D, .} up to length 8 were classified identically by the hand DFA, the Thompson-derived minimal DFA, the regular expression derived by state elimination and Python's re module (0 mismatches).

## 4.5 The pumping lemma and a non-regular clinical pattern

**Pumping lemma.** If L is regular there is a length $p \geq 1$ such that every string $s \in L$ with $|s| \geq p$ can be written $s = xyz$ with $|xy| \leq p$, $|y| \geq 1$, and $xy^iz \in L$ for every $i \geq 0$.

**Claim.** $L = \{ (^n )^n \mid n \geq 0 \}$ (n opening then n closing brackets, the simplest case of balanced nested brackets in a clinical note) is not regular.

**Proof.** Suppose L is regular with pumping length p. Take $s = (^p )^p \in L$, $|s| = 2p \geq p$. Since $|xy| \leq p$, both x and y lie inside the first p symbols, which are all "(", and $|y| \geq 1$, so $y = (^k$ with $k \geq 1$. Pumping up with i = 2 gives $xy^2z = (^{p+k} )^p$, which has more opening than closing brackets and so is not in L. This contradicts the lemma, so L is not regular. ■

**Relation to Task 3.** This is exactly the limit stated in 3.5: a regular layer can recognise a bracket pair but not arbitrary nesting. A context-free grammar (a pushdown automaton with a stack) is the next model needed, and the proof also explains why the regex tokenizer keeps brackets as single PUNCT tokens and makes no attempt to match them.

# TASK 5 – Inflectional Morphology (3 Marks)

## 5.1 Definitions and the inflection / derivation criteria

A **morpheme** is the smallest meaning-bearing unit. A **stem** is the part of a word to which affixes attach, and an **affix** is a bound morpheme (prefix or suffix). **Inflection** changes the form of a word to express grammatical information (number, tense, agreement, degree) without creating a new lexeme. A **paradigm** is the full set of inflected forms of one lexeme.

| Criterion | Inflection | Derivation |
|---|---|---|
| Category preservation | Keeps the part of speech (treat → treated: verb) | Often changes it (treat → treatment: verb → noun) |
| Productivity | Almost fully productive: every regular noun takes a plural | Selective and gappy (treatable but not *patientable) |
| Position relative to other affixes | Outermost: treat-ment-s | Inside inflection: *treat-s-ment is impossible |
| Meaning | Predictable grammatical meaning | May shift meaning (dress → dressing) |

## 5.2 Inflectional paradigms in the corpus, with five full paradigms

Attested feature counts in the corpus (44,539 tokens): A/BASE 4,547, N/PL 3,128, N/SG 7,916, V/3SG 533, V/BASE 1,132, V/PAST 2,515, V/PPART 1,018, V/PROG 573. Paradigms present: noun number (SG, PL; the possessive is in the lexicon but not generated in the synthetic corpus), verb base, third-person singular, past, past participle and progressive; adjective comparison (CMP, SUP) is in the lexicon for 20 gradable adjectives (largest, milder, hotter) but the corpus uses only the base form.

| Lexeme | Forms |
|---|---|
| patient/N | SG: patient · PL: patients · POSS: patient's |
| treat/V | BASE: treat · 3SG: treats · PAST: treated · PPART: treated · PROG: treating |
| diagnose/V | BASE: diagnose · 3SG: diagnoses · PAST: diagnosed · PPART: diagnosed · PROG: diagnosing |
| therapy/N | SG: therapy · PL: therapies · POSS: therapy's |
| plan/V | BASE: plan · 3SG: plans · PAST: planned · PPART: planned · PROG: planning |
| severe/A | BASE: severe · CMP: severer · SUP: severest |
| go/V | BASE: go · 3SG: gos · PAST: went · PPART: gone · PROG: going |

Allomorphs of the inflectional suffixes (23 realisations, more than the required 20):

| Category | Realisations |
|---|---|
| Noun plural / possessive (5) | -s, -es (after sibilant), -ies (after consonant + y), -’s, -s’ |
| Verb third person singular (3) | -s, -es, -ies |
| Verb past and past participle (5) | -ed, -d (after e), -ied, doubled consonant + -ed, -ked |
| Verb progressive (4) | -ing, doubled consonant + -ing, -king, -ying (lie → lying) |
| Adjective comparison (6) | -er, -est, -r, -st, -ier, -iest |

### 5.3 Orthographic rules, precisely, with an over-generating counter-example each

| Rule (a → b / c __ d) | Stated precisely | Over-generates when applied without its restriction |
|---|---|---|
| e-deletion e → ε / [^eoy] __ ^V | Delete a final e before a vowel-initial suffix (ed, ing, er, est) | Unrestricted: diagnose^s → "diagnoss" (correct: diagnoses) |
| consonant doubling C → CC / CVC~ __ ^V | Double the final consonant of a stem flagged with ~ (stress on the last syllable) before a vowel-initial suffix | Unflagged: visit^ed → "visitted" (correct: visited) |
| y → i y → i / C __ ^(ed\|er\|est); y → ie / C __ ^s | Consonant + y changes before these suffixes, not before -ing and not after a vowel | Without "consonant before": play^ed → "plaied" (correct: played) |
| sibilant e-insertion ε → e / (s\|x\|z\|ch\|sh)^ __ s | Insert e between a sibilant-final stem and -s | Over-generates in my own lexicon: stomach^s → "stomaches" (correct: stomachs), because ch is pronounced /k/ here |

### 5.4 Irregular forms, and why they are listed

The lexicon lists 66 irregular forms separately: nouns (diagnosis→diagnoses, prosthesis→prostheses, thrombus→thrombi, bacterium→bacteria, vertebra→vertebrae, fungus→fungi, …), verbs (be→was/been, have→had/had, do→did/done, go→went/gone, take→took/taken, give→gave/given, …) and adjectives (good → better, best; bad → worse, worst). They occur 1,100 times in the corpus (45 distinct forms). Each must be listed because no rule relates went to go, diagnoses to diagnosis, or thrombi to thrombus: the change is idiosyncratic, a rule that produced it would also produce wrong forms for regular words (an "-us → -i" rule would give *bi for bus), and these words are frequent enough that errors would be noticed.

### 5.5 Measured effect of inflection

**Type-to-lemma ratio** = distinct surface types / distinct lemmas = 1,035 / 441 = **2.35** (content words, 21,362 tokens). Each lemma therefore appears on average in about 2.3 surface forms, so an exact-match search sees more than twice as many "words" as there are concepts.

**Query failures explained by inflection.** With no normalisation, 470 of the 1,504 relevant (query, record) pairs are missed (31.3%). Resolving inflection with the transducer recovers 100.0% of those misses, so inflection alone accounts for all the misses in this synthetic setting. On real notes some misses would also come from derivation, misspelling and synonymy.

## TASK 6 – Derivational Morphology (4 Marks)

### 6.1 Definition and 24 attested derivational affixes

Derivation builds new lexemes by adding affixes, and typically changes the category (-ment: verb → noun) or the meaning (hyper-: above normal). The affixes below were counted in the corpus by type (distinct lemmas) and token. "Free base" is the number of types whose remaining stem is itself a word of the lexicon (treat + -ment).

| Affix | Category change | Types | Tokens | Free base | Examples |
|---|---|---|---|---|---|
| -itis | bound root → N (inflammation) | 5 | 169 | 0 | arthritis, bronchitis, gastritis |
| -ectomy | bound root → N (removal) | 3 | 104 | 0 | appendectomy, mastectomy, nephrectomy |
| -ology | bound root → N (study) | 6 | 201 | 0 | cardiology, nephrology, neurology |
| -ologist | bound root → N (specialist) | 6 | 221 | 0 | cardiologist, nephrologist, neurologist |
| -osis | bound root → N (condition) | 5 | 179 | 0 | cyanosis, diagnosis, fibrosis |
| -aemia | bound root → N (blood condition) | 3 | 113 | 0 | anaemia, hyperglycaemia, hypoglycaemia |
| -ment | V → N (result/process) | 8 | 324 | 3 | appointment, assessment, enlargement |
| -ness | A → N (state) | 3 | 103 | 2 | dizziness, tenderness, weakness |
| -ity | A → N (state) | 4 | 165 | 2 | mobility, sensitivity, stability |
| -able | V → A (can be …ed) | 3 | 235 | 2 | recoverable, stable, treatable |
| -ive | V → A (tending to) | 4 | 210 | 1 | hypertensive, operative, postoperative |
| -ous | N → A (characterised by) | 2 | 123 | 0 | conscious, infectious |
| -ic | N → A (pertaining to) | 11 | 551 | 0 | anaemic, analgesic, antibiotic |
| -al | N → A (pertaining to) | 16 | 1,010 | 0 | abdominal, abnormal, bilateral |
| -ion/-ation | V → N (action/result) | 15 | 530 | 6 | admission, complication, dysfunction |
| -ist | N → N (practitioner) | 7 | 257 | 6 | cardiologist, nephrologist, neurologist |
| hyper- | N/A → N/A (above normal) | 3 | 118 | 0 | hyperglycaemia, hypertension, hypertensive |
| hypo- | N/A → N/A (below normal) | 2 | 68 | 0 | hypoglycaemia, hypotension |
| post- | A → A (after) | 2 | 108 | 1 | posterior, postoperative |
| pre- | A → A (before) | 2 | 91 | 1 | preoperative, prescribe |
| nephr(o)- | bound → N (kidney) | 4 | 141 | 0 | nephrectomy, nephritis, nephrologist |
| dys- | N → N (abnormal) | 2 | 68 | 0 | dysfunction, dyspnoea |
| tachy-/brady- | N → N (fast/slow) | 2 | 80 | 0 | bradycardia, tachycardia |
| anti- | N → N (against) | 2 | 86 | 0 | antibiotic, antibody |

## 6.2 Derivational families, and impossible orders

| Root | Family (order of attachment) | Impossible order and why |
|---|---|---|
| nephr(o)- | nephr + -itis → nephritis; nephr + -ectomy → nephrectomy; nephr + -ology → nephrology + -ist → nephrologist | *nephr-ist-ology: -ist must follow -ology, because it forms a person from a field |
| infect | infect → infection (+ion) → infectious (+ous); infect + -ed (inflection) | *infect-ed-ion: inflection (-ed) can never sit inside a derivational suffix |
| operate | operate → operation → operational; operate → operative → postoperative (post- attaches last) | *operative-post (prefix after the stem), *operation-ate |
| treat | treat → treatment → treatments; treat → treatable | *treat-s-ment: plural inside the derivation |
| cardi(o)- | cardi + -ac; cardi + -ology → cardiology → cardiologist | *cardio-ist-logy: the specialist suffix needs the field suffix first |
| hyper-<br>tens- | tens + -ion → tension; hyper- + tension → hypertension; + -ive → hypertensive | *tension-hyper (prefix must precede); hyper-hypo- never stack |

## 6.3 Non-compositional meaning and what it means for aggressive stemming

The meaning of a derived word is often not the sum of its parts. In this lexicon: **dressing** (a wound covering) is not "dress + -ing"; **stable** (unchanging) is not st + -able, and a naive "-able" rule counted it among the three -able types in the table above; **discharge** (leaving hospital) is not the opposite of "charge"; **operation** means a surgical procedure, not any act of operating; **conscious** is not con + scious in a way a reader can use. An aggressive stemmer therefore creates false families. The measured result: Porter stemming merges 43 of 443 lemmas into 21 shared stems (e.g. infect/infection; inject/injection; infuse/infusion; dress/dressing; pain/painful; swell/swelling). A search system should use affix rules only together with the lexicon, and must not strip prefixes blindly: stripping hyper-/hypo- would turn hyperglycaemia and hypoglycaemia into the same word.

## 6.4 Productivity of the affixes

Productivity was classified by the evidence available in this corpus: number of types, and the share of types built on a free base (a live process attaches to free-standing words; a fossilised one only occurs on bound roots). I did **not** use the hapax-based productivity measure, because every lemma in a 500-record synthetic corpus occurs many times and there are no hapax legomena to count.

| Affix | Types | Free-base types | Free-base share | Classification |
|---|---|---|---|---|
| -itis | 5 | 0 | 0.00 | bound roots only: fossilised in this lexicon |
| -ectomy | 3 | 0 | 0.00 | bound roots only: fossilised in this lexicon |
| -ology | 6 | 0 | 0.00 | bound roots only: fossilised in this lexicon |
| -ologist | 6 | 0 | 0.00 | bound roots only: fossilised in this lexicon |
| -osis | 5 | 0 | 0.00 | bound roots only: fossilised in this lexicon |

| Affix | Types | Free-base types | Free-base share | Classification |
|---|---|---|---|---|
| -aemia | 3 | 0 | 0.00 | bound roots only: fossilised in this lexicon |
| -ment | 8 | 3 | 0.38 | productive (free bases) |
| -ness | 3 | 2 | 0.67 | productive (free bases) |
| -ity | 4 | 2 | 0.50 | productive (free bases) |
| -able | 3 | 2 | 0.67 | productive (free bases) |
| -ive | 4 | 1 | 0.25 | bound roots only: fossilised in this lexicon |
| -ous | 2 | 0 | 0.00 | bound roots only: fossilised in this lexicon |
| -ic | 11 | 0 | 0.00 | bound roots only: fossilised in this lexicon |
| -al | 16 | 0 | 0.00 | bound roots only: fossilised in this lexicon |
| -ion/-ation | 15 | 6 | 0.40 | productive (free bases) |
| -ist | 7 | 6 | 0.86 | productive (free bases) |
| hyper- | 3 | 0 | 0.00 | bound roots only: fossilised in this lexicon |
| hypo- | 2 | 0 | 0.00 | bound roots only: fossilised in this lexicon |
| post- | 2 | 1 | 0.50 | bound roots only: fossilised in this lexicon |
| pre- | 2 | 1 | 0.50 | bound roots only: fossilised in this lexicon |
| nephr(o)- | 4 | 0 | 0.00 | bound roots only: fossilised in this lexicon |
| dys- | 2 | 0 | 0.00 | bound roots only: fossilised in this lexicon |
| tachy-/brady- | 2 | 0 | 0.00 | bound roots only: fossilised in this lexicon |
| anti- | 2 | 0 | 0.00 | bound roots only: fossilised in this lexicon |

Caveat: -itis and -ectomy are classified “fossilised” only relative to this lexicon, because they attach to bound Greek roots. In real clinical English they are very productive (new -itis terms are coined easily); the table says that a **free-base rule** would not find them, so they must be handled by listing or by bound-root lists, not by stripping.

## 6.5 Should derivational relatives be conflated for search?

**Recommendation: not by default; offer it as an optional query expansion.** Cases where conflation **helps** the clinician: (1) a search for infection should also find infect-ed and infectious; (2) examine/examination, investigate/investigation and assess/assessment describe the same activity. Cases where conflation is **clinically misleading**, taken from the measured Porter collisions: (1)

**hepatic** (liver-related) and **hepatitis** (a disease) collapse to the same stem "hepat"; (2) **dress** and **dressing** (a wound covering) collapse to "dress", and a search for the verb returns wound-care notes. A second pair of dangers is the prefix pair hyper-/hypo- discussed above. The lemma search of Task 9 never conflates across derivation, which is why its precision stays high.

## TASK 7 – Finite-State Morphological Parsing (5 Marks)

### 7.1 A transducer as a two-tape automaton

A finite-state transducer (FST) is a finite automaton whose arcs carry pairs a:b, reading symbol a on one tape and writing b on the other. Here the **lexical tape** holds a stem and features (fracture +N +PL) and the **surface tape** holds the spelled word (fractures). The same pair of tapes can be read in either direction.

**Worked example.** Lexical fracture +N +PL; the lexicon FST maps stem letters to themselves (f:f r:r a:a c:c t:t u:u r:r e:e), then the arc +N:ε (no surface output) and +PL:^s writes the intermediate boundary and suffix, giving fracture^s. The rule transducers (Task 8) leave this unchanged, and deleting the boundary ^ yields the surface tape fractures. Run backwards, reading surface fractures consumes the same arcs in reverse and emits fracture +N +PL.

### 7.2 The lexicon as continuation classes

| Class | Used for | Suffix arcs (lexical : surface) | Stems |
|---|---|---|---|
| N_REG | regular nouns | +N +SG : ε ; +N +PL : ^s ; +N +POSS : ^'s | patient, tablet, therapy, … |
| N_MASS | mass nouns (no plural) | +N +SG : ε | blood, oxygen, diabetes, … |
| N_IRR | irregular nouns | +N +PL : listed form | diagnosis→diagnoses, child→children, … |
| V_REG | regular verbs | +V +BASE : ε ; +3SG : ^s ; +PAST, +PPART : ^ed ; +PROG : ^ing | treat, diagnose, … |
| V_DBL | verbs with stressed final CVC | as V_REG, with the doubling flag ~ on the stem | plan~, scan~, admit~, refer~, stop~ |
| V_IRR | irregular verbs | +PAST, +PPART : listed ; +3SG, +PROG : regular | go→went/gone, take→took/taken, … |
| A_REG / A_IRR | adjectives | +A +BASE : ε ; +CMP : ^er ; +SUP : ^est (gradable only) | severe, mild, large; good→better/best |
| DER | derived words | listed entries with a derivation link to the base (Task 6) | treatment ← treat + -ment, … |

The lexicon covers three parts of speech (noun, verb, adjective) with 463 entries and compiles to a relation of 1,520 (lexical, surface) pairs. Derivational affixes are handled as listed derived entries linked to their bases rather than generated productively, because of the non-compositionality shown in Task 6.

## 7.3 Hand-traced parses (state sequences and tapes)

| Surface (read) | State sequence in the lexicon FST | Intermediate string | Rules fired | Analysis (written) |
|---|---|---|---|---|
| fractures | q0 → s[f] → s[fr] → s[fra] → s[frac] → s[fract] → s[fractu] → s[fractur] → s[fracture] → C:N_REG → suf[s] | fracture^s | none | fracture +N +PL |
| treated | q0 → s[t] → s[tr] → s[tre] → s[trea] → s[treat] → C:V_REG → suf[e] → suf[ed] | treat^ed | none | treat +V +PAST |
| planned | q0 → s[p] → s[pl] → s[pla] → s[plan] → C:V_DBL → suf[e] → suf[ed] | plan~^ed | R2 doubling | plan +V +PAST |
| therapies | q0 → s[t] → s[th] → s[the] → s[ther] → s[thera] → s[therap] → s[therapy] → C:N_REG → suf[s] | therapy^s | R4 y->ie | therapy +N +PL |
| diagnosing | q0 → s[d] → s[di] → s[dia] → s[diag] → s[diagn] → s[diagno] → s[diagnos] → s[diagnose] → C:V_REG → suf[i] → suf[in] → suf[ing] | diagnose^ing | R3 e-deletion | diagnose +V +PROG |

Reading the first row: the machine walks the stem letters f-r-a-c-t-u-r-e, reaches the continuation class N_REG, then takes the suffix arc s, which on the lexical tape is +N +PL; the surface symbols consumed are fractures and the lexical symbols emitted are fracture +N +PL. For “planned” the doubling rule R2 fires and the parser undoes it, recovering plan~^ed; for “therapies” rule R4 (y → ie before ^s) fires.

## 7.4 Generation with the same machine

Parsing and generation read one and the same relation in opposite directions. The relation is compiled once from lexicon and rules; parse(s) looks it up on the surface side and gen(l) on the lexical side, and there is no second machine. Check: for all 1,520 (lexical, surface) pairs, parsing the generated surface form returned the original lexical analysis (1,520 of 1,520).

| Lexical input | Generated surface form |
|---|---|
| fracture +N +PL | fractures |
| treat +V +PAST | treated |
| plan +V +PROG | planning |
| therapy +N +PL | therapies |
| go +V +PAST | went |

## 7.5 Ambiguity

| Surface form | All analyses returned |
|---|---|
| records | record +N +PL  \|  record +V +3SG |

| Surface form | All analyses returned |
|---|---|
| diagnoses | diagnosis +N +PL \| diagnose +V +3SG |
| fractures | fracture +N +PL \| fracture +V +3SG |
| scans | scan +N +PL \| scan +V +3SG |
| burns | burn +N +PL \| burn +V +3SG |

The transducer returns all analyses; 178 of 1,426 surface forms are ambiguous. **How a downstream component chooses:** a part-of-speech tagger that looks at the preceding word. My simple rule version (after the/a/of/no prefer noun; after to prefer the base verb; after was/were prefer the participle; otherwise prefer verb) resolved 3,427 of 4,055 ambiguous tokens correctly (84.5%). A statistical tagger trained on annotated notes would replace those rules; for search, indexing all analyses and accepting some false positives is also a defensible choice (Task 9).

## TASK 8 – Orthographic Rules and Two-Level Morphology (4 Marks)

### 8.1 Why the lexicon transducer alone gives wrong spellings

| Lexical form | Plain concatenation | Correct surface |
|---|---|---|
| plan + ed | planed | planned |
| diagnose + ing | diagnoseing | diagnosing |
| therapy + s | therapys | therapies |
| abscess + s | abscesss | abscesses |
| panic + ed | paniced | panicked |

A lexicon FST concatenates stem and suffix. Spelling changes depend on the letters on both sides of the boundary, which the lexicon does not see, so a second layer of rules is needed. The alternative of listing every surface form in the lexicon would multiply the lexicon by about four.

### 8.2 Seven rewrite rules (a → b / c __ d), ^ = morpheme boundary, ~ = doubling flag

| Rule | Notation | Example |
|---|---|---|
| R1 ie → y | ie → y / __ ^ing | lie^ing → lying |
| R2 consonant doubling | C → CC / [^aeiou][aeiou] __ ~^V (flagged stems only) | plan~^ed → planned |
| R3 e-deletion | e → ε / [^eoy] __ ^V | diagnose^ing → diagnosing; die^ed → died |
| R4 y → ie | y → ie / C __ ^s | therapy^s → therapies |
| R5 y → i | y → i / C __ ^(ed\|er\|est) | dry^ed → dried |
| R6 k-insertion | ε → k / ic __ ^(ed\|ing\|er) | panic^ed → panicked |
| R7 e-insertion | ε → e / (s\|x\|z\|ch\|sh)^ __ s | abscess^s → abscesses |

## 8.3 Composition and the cascade

Each rewrite rule is a regular relation and is therefore implemented as a transducer. Composing them in order with the lexicon FST gives one transducer from lexical strings to surface strings. Order dependencies found: **R1 must precede R3** (die^ing: if e-deletion ran first it would give “diing”; R1 first gives dy^ing → dying); **R2 must precede the flag deletion and R3**, because it needs the ~ marker; **R4/R5 precede R7** in design so that a y-stem never receives the sibilant rule (they do not interact for the test set but the order is kept fixed); R6 and R7 act on different contexts and are order-independent.

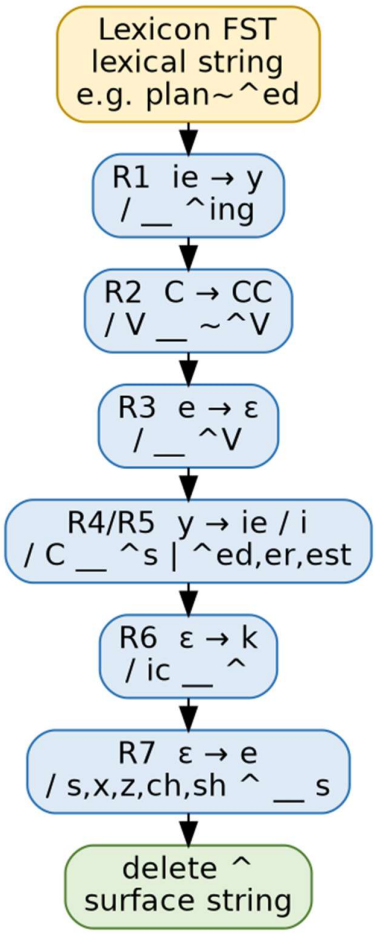



*Figure 6. Rule cascade composed with the lexicon transducer.*

## 8.4 Cascade versus two-level (Koskenniemi) morphology

The **cascade** applies rules one after another to intermediate strings, which is easy to write and read and is how I implemented it, but intermediate forms must be kept consistent and the order is delicate (R1 before R3 above). **Two-level** rules are constraints on aligned lexical:surface character pairs that are applied in parallel with no intermediate level (e.g. lexical e may correspond to surface 0 only when followed by ^ and then ing/ed). Advantages of two-level: no ordering problems, rules are declarative and symmetric, and one rule set works equally for parsing and generation. Advantages of the cascade: rules are small and familiar, each can be tested alone, and a cascade composes into a single transducer with a standard algorithm, which makes the implementation here straightforward. I implemented the cascade only; the two-level form is described, not built.

## 8.5 Test of the complete system

32 lexical inputs exercising every rule (and the irregular list) were generated and compared with the correct spelling: **29 correct, 3 failures.**

| Lexical input | Rule exercised | Generated | Correct | Result |
|---|---|---|---|---|
| diagnose +PROG | R3 | diagnosing | diagnosing | ✓ |
| manage +PAST | R3 | managed | managed | ✓ |
| relieve +PROG | R3 | relieving | relieving | ✓ |
| cannulate +PAST | R3 | cannulated | cannulated | ✓ |
| large +SUP | R3 | largest | largest | ✓ |
| plan +PAST | R2 | planned | planned | ✓ |
| scan +PROG | R2 | scanning | scanning | ✓ |
| admit +PAST | R2 | admitted | admitted | ✓ |
| refer +PROG | R2 | referring | referring | ✓ |
| stop +PAST | R2 | stopped | stopped | ✓ |
| occur +PAST | R2 | occurred | occurred | ✓ |
| visit +PAST | R2 (blocked) | visited | visited | ✓ |
| monitor +PROG | R2 (blocked) | monitoring | monitoring | ✓ |
| therapy +PL | R4 | therapies | therapies | ✓ |
| biopsy +PL | R4 | biopsies | biopsies | ✓ |
| dry +PAST | R4/R5 | dried | dried | ✓ |
| dry +3SG | R4/R5 | dries | dries | ✓ |
| dry +PROG | R4/R5 | drying | drying | ✓ |
| panic +PAST | R6 | panicked | panicked | ✓ |
| panic +PROG | R6 | panicking | panicking | ✓ |
| abscess +PL | R7 | abscesses | abscesses | ✓ |
| pass +3SG | R7 | passes | passes | ✓ |
| fix +3SG | R7 | fixes | fixes | ✓ |
| wash +3SG | R7 | washes | washes | ✓ |
| lie +PROG | R1 | lying | lying | ✓ |
| die +PAST | R3 | died | died | ✓ |
| stomach +PL | R7 | stomaches | stomachs | ✗ |
| hot +CMP | R2 | hoter | hotter | ✗ |
| fit +CMP | R2 | (no form) | fitter | ✗ |
| go +PAST | list | went | went | ✓ |
| diagnosis +PL | list | diagnoses | diagnoses | ✓ |
| take +PPART | list | taken | taken | ✓ |

**Account of every failure:** (1) **stomach + PL → "stomaches"**: rule R7 treats "ch" as a sibilant, but here ch is pronounced /k/ and the correct plural is stomachs; this is a genuine over-generation, fixed by a lexical exception flag. (2) **hot + CMP → "hoter"**: the doubling flag was implemented for verbs only; the adjective hot needs the ~ flag in the lexicon. This is a lexicon-data gap, not a rule error. (3) **fit + CMP → no form**: the adjective fit was not marked as gradable (CMP) in the lexicon, so no comparative is generated; again a data gap. No rule was wrong in cases (2) and (3), which shows that the cost of a finite-state morphology is dominated by lexicon maintenance.

## TASK 9 – The Porter Stemmer (5 Marks)

### 9.1 The algorithm

**Consonants and vowels.** A letter is a vowel if it is a, e, i, o or u, or y preceded by a consonant; otherwise it is a consonant. Any word then has the form $[C](VC)^m[V]$, where C and V are maximal runs of consonants and vowels. The **measure m** is the number of VC repetitions. **Conditions:** *v* = the stem contains a vowel; *d = the stem ends in a double consonant; *o = the stem ends in consonant-vowel-consonant where the final consonant is not w, x or y.

| Word | C/V pattern | Collapsed | m |
|---|---|---|---|
| tree | CCVV | CV | 0 |
| trouble | CCVVCCV | CVCV | 1 |
| oats | VVCC | VC | 1 |
| treat | CCVVC | CVC | 1 |
| diagnosis | CVVCCVCVC | CVCVCVC | 3 |
| hypertension | CVCVCCVCCVVC | CVCVCVCVC | 4 |
| infection | VCCVCCVVC | VCVCVC | 3 |
| nephrectomy | CVCCCVCCVCV | CVCVCVCV | 3 |

Reading one row: diagnosis = CVCVCVCVC with runs collapsed to C V C V C V C → [C](VC)(VC)(VC): m = 3. The five steps: **1** plurals and -ed/-ing (1a, 1b, 1c), **2** double suffixes mapped to single (ational → ate), **3** -icate, -ative, -ful, -ness, **4** removal of -al, -ance, -ent, -ion, -ive … when m > 1, **5** final -e and -ll tidy-up.

### 9.2 Hand traces through the five steps (ten clinical words)

| Word | m | 1a | 1b | 1c | 2 | 3 | 4 | 5a | 5b | Stem |
|---|---|---|---|---|---|---|---|---|---|---|
| infections | 3 | **infection** | · | · | · | · | **infect** | · | · | infect |
| nephrectomies | 4 | **nephrectomi** | · | · | · | · | · | · | · | nephrectomi |
| hypertension | 4 | · | · | · | · | · | **hypertens** | · | · | hypertens |
| diagnosing | 3 | · | **diagnos** | · | · | · | · | · | · | diagnos |
| treatment | 2 | · | · | · | · | · | · | · | · | treatment |
| operational | 5 | · | · | · | **operate** | · | **oper** | · | · | oper |
| hospitalization | 6 | · | · | · | **hospitalize** | **hospital** | **hospit** | · | · | hospit |
| effectiveness | 5 | **effectiveness** | · | · | **effective** | · | **effect** | · | · | effect |

| Word | m | 1a | 1b | 1c | 2 | 3 | 4 | 5a | 5b | Stem |
|---|---|---|---|---|---|---|---|---|---|---|
| recoverable | 4 | · | · | · | · | · | **recover** | · | · | recover |
| dressings | 2 | **dressing** | **dress** | · | · | · | · | · | · | dress |

A bold form shows a step that changed the word; "·" means no rule fired. The rules that fired, step by step:

| Word | Step | Before → after | Rule |
|---|---|---|---|
| infections | 1a | infections → infection | S -> (none) |
| infections | 4 | infection → infect | ION -> (none) (m>1) |
| nephrectomies | 1a | nephrectomies → nephrectomi | IES -> I |
| hypertension | 4 | hypertension → hypertens | ION -> (none) (m>1) |
| diagnosing | 1b | diagnosing → diagnos | ING -> (none) (*v*) |
| operational | 2 | operational → operate | ATIONAL -> ATE (m>0) |
| operational | 4 | operate → oper | ATE -> (none) (m>1) |
| hospitalization | 2 | hospitalization → hospitalize | IZATION -> IZE (m>0) |
| hospitalization | 3 | hospitalize → hospital | ALIZE -> AL (m>0) |
| hospitalization | 4 | hospital → hospit | AL -> (none) (m>1) |
| effectiveness | 1a | effectiveness → effectiveness | SS -> SS |
| effectiveness | 2 | effectiveness → effective | IVENESS -> IVE (m>0) |
| effectiveness | 4 | effective → effect | IVE -> (none) (m>1) |
| recoverable | 4 | recoverable → recover | ABLE -> (none) (m>1) |
| dressings | 1a | dressings → dressing | S -> (none) |
| dressings | 1b | dressing → dress | ING -> (none) (*v*) |

The three conditions in action: **plastered** → plaster (ED -> (none) (*v*)); **hopping** → hop (ING -> (none) (*v*); undouble (*d, not L/S/Z)); **filing** → file (ING -> (none) (*v*); add E (m=1, *o)). Also instructive: **treatment** is *not* stemmed (m of "treat" is 1, and step 4 needs m > 1), so treat and treatment stay apart: an under-stemming example.

**Conformance:** the implementation reproduces 63 of 63 published example pairs from the original description.

## 9.3 Stemming versus lemmatisation on 200 annotated forms

The 200 forms are 200 distinct (surface, lemma) pairs sampled from the corpus, 30 of them irregular. For each, I tested whether the Porter stem of the surface equals the Porter stem of the gold lemma ("conflates") and whether the transducer's lemma set contains the gold lemma.

| Outcome | Count | Meaning |
|---|---|---|
| Both succeed | 167 | stem and lemma both relate the form to its lemma |
| Stemmer fails, lemmatiser succeeds | 33 | under-stemming: fungi/fungus, took/take, brought/bring, went/go, drank/drink |

| Outcome | Count | Meaning |
|---|---|---|
| Lemmatiser fails, stemmer succeeds | 0 | none (the lemmatiser's lexicon is the generator's: circular, see the note at the start) |
| Both fail | 0 | – |

They diverge only on irregular forms and a few -us/-is nouns (fungi/fungus, diagnoses/diagnosis): a rule stemmer cannot see them. Note the caveat: the lemmatiser looks perfect here only because the corpus was generated from its own lexicon.

## 9.4 Over-stemming and under-stemming

**Under-stemming** (related forms not conflated): 33 of 200 annotated forms = **16.5%**, almost all irregulars (went, took, fungi, brought). **Over-stemming** (distinct lemmas sharing a stem): 18 of 200 annotated forms (9.0%) have a stem shared with another lexicon lemma; over the whole lexicon 43 of 443 lemmas are involved (9.7%) in 21 collisions. Clinically significant ones: hepat: hepatic/hepatitis; dress: dress/dressing; diabet: diabetes/diabetic; oper: operate/operation/operative.

## 9.5 Retrieval effect

Forty queries (the citation form of a lemma, six of them irregular such as go, drink, fungus, bacterium) were run over the 500 records under three conditions. Relevance = record contains any form of the lemma.

| Condition | Precision | Recall | F1 | Index terms | TP / FP / FN |
|---|---|---|---|---|---|
| no normalisation | 1.000 | 0.688 | 0.815 | 1,120 | 1034 / 0 / 470 |
| Porter stemmer | 0.834 | 0.958 | 0.892 | 556 | 1441 / 287 / 63 |
| finite-state analysis | 0.974 | 1.000 | 0.987 | 527 | 1504 / 40 / 0 |

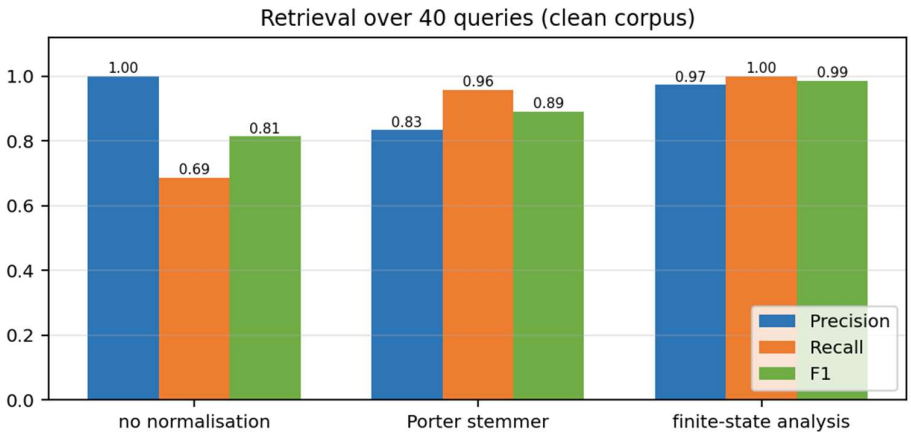



*Figure 7. Retrieval quality under the three conditions (clean corpus).*

No normalisation has perfect precision but misses 31.3% of the relevant records. Porter stemming recovers most of them (recall 0.96) but pays in precision (0.83) because of the collisions above. Full morphological analysis reaches recall 1.00 with precision 0.97; its 40 false positives come from the ambiguity of words such as diagnoses (noun or verb) when all analyses are indexed. **Again the finite-state figures are a ceiling** (see the note at the start).

## TASK 10 – Implement and Test the Pipeline (3 Marks)

### 10.1 Implementation

All components are implemented in Python using the standard library (re, collections, time, json); Matplotlib is used only for plots. NLTK was not installed in the working environment, so the Porter stemmer is my own implementation, validated against published pairs (Task 9). Modules: porter.py, automata.py (DFA, NFA, subset construction, minimisation, Thompson construction, state elimination), morph.py (lexicon, rule cascade, transducer relation), pipeline.py (patterns, tokenizer, normaliser), corpus.py (generator), dfas.py, and the experiment scripts run1–run3.py.

Subset construction as implemented (excerpt):

```
def subset_construction(nfa):
    s0 = nfa.eclose({nfa.start}); names = {s0: 'A'}; order = [s0]; rows = []; delta = {}
    i = 0
    while i < len(order):
        S = order[i]; i += 1
        for a in nfa.alphabet:
            mv = nfa.move(S, a)
            if not mv: continue
            T = nfa.eclose(mv)                # epsilon closure of the move
            if T not in names:
                names[T] = chr(ord('A') + len(names)); order.append(T)
            delta[(names[S], a)] = names[T]
            rows.append((names[S], sorted(S), a, sorted(mv), sorted(T), names[T]))
    final = {names[S] for S in order if S & nfa.final}
    return DFA(delta, 'A', final, nfa.alphabet), rows, names
```

Morphological generation through the rule cascade (excerpt):

```
RULES = [('R1 ie->y', r'ie(?=\^ing)', 'y'),
 ('R2 doubling', r'(?<=[^aeiou][aeiou])([bdglmnprstz])~(?=\^[aeiou])', r'\1\1'),
 ('R0 strip flag', r'~', ''),
 ('R3 e-deletion', r'(?<=[^eoy])e(?=\^[aeiou])', ''),
 ('R4 y->ie', r'(?<=[^aeiou])y(?=\^s$)', 'ie'),
 ('R5 y->i', r'(?<=[^aeiou])y(?=\^(?:ed|er|est)$)', 'i'),
 ('R6 k-insertion', r'(?<=ic)(?=\^(?:ed|ing|er)$)', 'k'),
 ('R7 e-insertion', r'(s|x|z|ch|sh)(?=\^s$)', r'\1e')]
def cascade(lexical):
    s = lexical
    for name, pat, rep in RULES: s = re.sub(pat, rep, s)
    return s.replace('^', '')
```

### 10.2 Stage-by-stage results over the corpus

| Stage | Items in | Output / accepted | Rejected | Run time (ms) | Unit |
|---|---|---|---|---|---|
| Tokenizer (maximal munch) | 500 | 53,954 | 0 | 144.9 | records→tokens |
| Entity regexes (13 patterns) | 500 | 5,380 | 0 | 85.4 | records→matches |
| DFA: patient ID validator | 477 | 415 | 62 | 0.6 | candidates→accepted |
| DFA: ISO date validator | 233 | 44 | 189 | 0.2 | candidates→accepted |
| DFA: blood-pressure validator | 325 | 325 | 0 | 0.5 | candidates→accepted |
| DFA: ICD validator | 306 | 306 | 0 | 0.2 | candidates→accepted |

| Stage | Items in | Output / accepted | Rejected | Run time (ms) | Unit |
|---|---|---|---|---|---|
| Normaliser | 53,954 | 44,539 | 9,415 | 157.4 | tokens→tokens (punctuation dropped) |
| FST morphological analyser | 23,731 | 23,397 | 334 | 5.2 | open-class tokens→analysed |
| Porter stemmer | 39,211 | 39,211 | 0 | 343.6 | tokens→stems |

Reading the table: the DFA validators reject strings that the permissive regular expressions accepted but that fall outside the strict canonical format (for example only 44 of 233 date candidates are in ISO form, the rest use dd-mm-yyyy and similar); the analyser rejects 334 open-class tokens as out-of-lexicon, a coverage of 98.6%.

## 10.3 Test cases checked against hand-computed results

| Test | Result |
|---|---|
| DFA patient_id: accepts 4 valid strings | pass |
| DFA patient_id: rejects 4 invalid strings | pass |
| DFA iso_date: accepts 2 valid strings | pass |
| DFA iso_date: rejects 3 invalid strings | pass |
| DFA icd: accepts 3 valid strings | pass |
| DFA icd: rejects 4 invalid strings | pass |
| DFA bp: accepts 3 valid strings | pass |
| DFA bp: rejects 4 invalid strings | pass |
| NFA ≡ subset-DFA ≡ minimised DFA on 6-symbol strings | pass |
| Thompson NFA ≡ hand DFA ≡ derived regex ≡ Python re (9,841 strings) | pass |
| FST parse fractures → fracture… (all analyses) | pass |
| FST parse treated → treat… (all analyses) | pass |
| FST parse diagnoses → diagnose… (all analyses) | pass |
| FST parse went → go… (all analyses) | pass |
| FST parse children → child… (all analyses) | pass |
| FST parse records → record… (all analyses) | pass |
| FST round trip parse(gen(x)) ∋ x for all relation pairs | pass |
| Porter: 63 published test pairs reproduce | pass |
| Regex negatives: 'BP 120/80' without unit not matched by bp pattern | pass |
| Regex negatives: 'Pt-123456' not matched by patient_id | pass |
| Regex negatives: textual date not matched by date pattern | pass |
| Regex negatives: 5 mg alone not a drug+strength | pass |

All 22 checks pass. They include at least one rejected string per automaton, the ambiguous forms of Task 7 (fractures, treated, diagnoses, records), the irregulars of Task 5 (went, children) and agreement with the hand results of Tasks 4, 7 and 9.

### 10.4 Plots

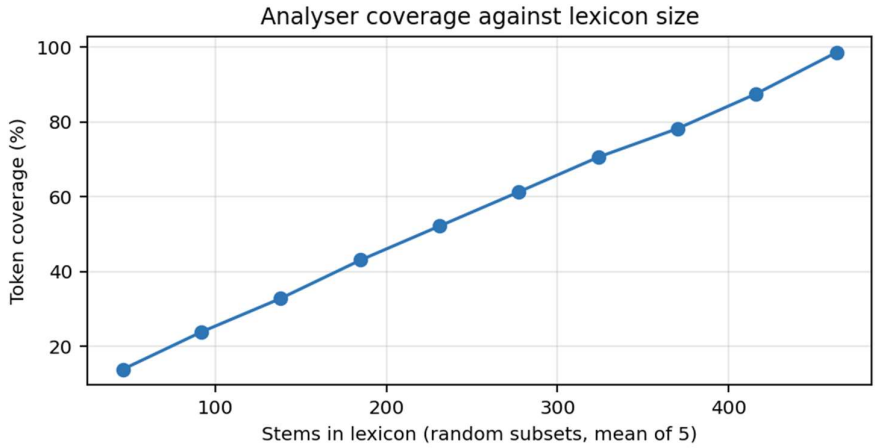



*Figure 8. Token coverage of the analyser as the lexicon grows (random subsets of the 463 lexicon entries, mean of five draws).*

Coverage rises roughly linearly (about 0.23 percentage points per added stem at the top end) and reaches 98.6% with the full lexicon: no knee appears, so every additional hand-built entry still pays.

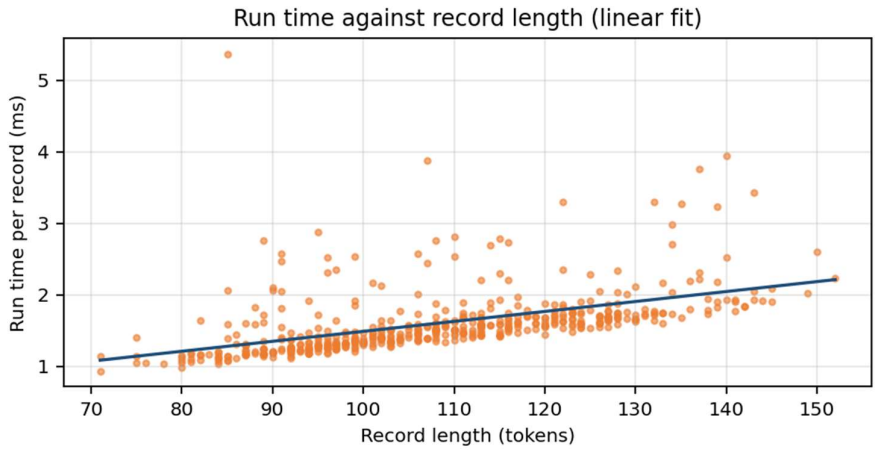



*Figure 9. Run time per record against record length.*

The fitted slope is 0.0139 ms per token (r = 0.50): the pipeline is linear in record length, but the correlation is weak because record type changes the number of entities and regex matches, not only the token count.

## TASK 11 – Compare, Recommend and Defend the Normalisation Design (2 Marks)

### 11.1 Comparison on measured results

| Criterion | No normalisation | Porter stemming | Finite-state analysis |
|---|---|---|---|
| Retrieval P / R / F1 | 1.00 / 0.69 / 0.81 | 0.83 / 0.96 / 0.89 | 0.97 / 1.00 / 0.99 (ceiling) |
| Coverage of tokens | 100% (every token is its own key) | 100% (always produces a stem) | 98.6% (unknown words get no analysis) |

| Criterion | No normalisation | Porter stemming | Finite-state analysis |
|---|---|---|---|
| Run time (this corpus) | ≈0 | 344 ms for 39,211 tokens (8.8 ms per 1000) | 5.2 ms for 23,731 tokens (0.22 ms per 1000, one dictionary lookup) |
| Handles irregular forms | No | No (under-stems 16.5% of sample) | Yes, if listed |
| Maintenance effort (my estimate) | none | ≈1 h (fixed algorithm, add test words) | ≈45 h to build lexicon, rules and annotation (see 11.3) plus ongoing lexicon upkeep |

The finite-state analyser is faster here than the stemmer only because it is a table lookup and my Porter implementation is unoptimised Python; run time is not a deciding factor in any of the three.

## 11.2 Recommendation

**Deploy a hybrid: the finite-state analyser as the primary conflation, with the Porter stemmer as fallback for out-of-lexicon tokens, and no derivational conflation by default.** Expected accuracy on real notes: not the 0.99 measured here. Because the ceiling comes from a generated corpus, a realistic expectation is F1 between the Porter figure (0.89) and the ceiling, to be fixed by a clinician-labelled sample before go-live. Day-4 results (Task 12) give the best evidence of how much performance falls when the vocabulary is not closed.

**Failure modes and what a clinician would notice:** (1) a drug name or eponym not in the lexicon falls back to the stemmer, so related forms may not be found ("I searched for the drug and got only exact matches"); (2) ambiguous words such as diagnoses return a few extra records (a noun/verb confusion; mild noise); (3) a missing lexicon flag (hot → hotter) means one form is not found and nothing warns the user; (4) the system cannot see negation, so a search for pneumonia also returns "no evidence of pneumonia", which a clinician would notice at once and which no morphology can fix.

## 11.3 Manual effort and when learned methods become cheaper

| Activity | Quantity | Estimated person-hours |
|---|---|---|
| Writing and testing entity patterns | 13 patterns, error analysis | 10 |
| Designing and checking automata | 5 automata, tables, proofs | 10 |
| Building the lexicon | 461 stems at ≈4 minutes each | 31 |
| Listing irregular forms and rules | 66 irregular forms, 7 rules | 6 |
| Annotation of 200 forms and 40 query judgements | 240 judgements | 8 |
| Building the evaluation corpus | documented generator | 6 |
| **Total (estimate)** | | **≈ 71** |

The coverage curve in Figure 8 is roughly linear, so the cost of hand-building rises in proportion to vocabulary size. Hand-built resources are the more economical choice while the needed vocabulary is some hundreds to a few thousand lexemes and the text is clean. They stop being economical when the vocabulary is open-ended (drug brand names, eponyms, abbreviations, misspellings, several languages): the long tail then costs hand-built effort in proportion to its size, whereas a learned

model amortises that cost over annotated examples. My estimate, not a measurement, is that the crossover lies in the low thousands of lexemes or when more than about a fifth of tokens are out of the lexicon (which Day 4 expects); that is the reason the recommended design keeps a stemmer fallback and a plan for a learned tagger and lemmatiser.

## TASK 12 – Day-4 Design Challenge: Re-evaluate and Adapt the Design (8 Marks)

Changed requirement: OCR output from scanned handwritten forms (O/0, l/1, rn/m confusions, hyphenated line breaks, lost diacritics) and Romanised regional-language notes mixed with English (e.g. “patient ku fever irundhuchu, paracetamol 500mg kuduthen”), with at least 20% of tokens out of lexicon, misspelt or non-English.

### How the noisy sample was made (so the results can be judged)

- 300 records drawn at random from the 500: 100 with **OCR noise only**, 100 **code-mixed only**, 100 **both**.
- OCR noise: each entity string is corrupted with probability 0.35 and each ordinary word with probability 0.12, by one random confusion from {o↔0, l↔1, I→l, m→rn, B→8, S↔5}; words of eight or more letters are broken across a line end with probability 0.05. Diacritics do not occur in the English source text, so diacritic loss is not simulated.
- Code-mixing: after each word a Romanised Tamil or Hindi token is inserted with probability 0.30 (25 regional lexemes, three spellings each; one spelling per word is held out and never given to the system). Result: regional tokens are 19.8% of all word tokens in the code-mixed and combined groups, and 42.5% of the open-class tokens of the combined group are out of the original lexicon.
- Caveat: this is simulated noise with simple confusion models, and a real OCR engine produces other errors (merged words, missing characters, whole-line failures). The results show the *direction and rough size* of the problem, not exact production figures.

### A. Impact analysis (2 marks)

#### A.1 What the new input contains that the old design cannot handle

| Property of the new input | Component that fails | Example |
|---|---|---|
| Character-level noise | Patterns and automata (exact match) | PAT523S72 (S for 5), “500 rng”, “12O/8O”, “O.5 ml” |
| Words broken at line ends | Tokeniser: one word becomes two unknown tokens | “frac-⏎ture” |
| Out-of-lexicon stems | Analyser: no analysis at all | “Patlent”, “cornpleted”, “Fe1t” |
| Non-English morphology | Analyser and Porter: rules written for English suffixes mis-segment | “irundhuchu”, “kuduthen”, “dard” |
| Token-level code-mixing and spelling variants | Dictionary lookup, search index | irundhuchu / irunthuchu / irundhuchi are three index terms for one word |

#### A.2 Assumptions that no longer hold

- **Clean typed input:** the patterns assume exact characters.

- **A closed vocabulary:** the analyser either knows a word or knows nothing; at least 20% of tokens now fall outside.
- **English-only morphology:** the suffix rules and Porter's rules are English; applying them to Tamil or Hindi words gives meaningless "stems".
- **Exact acceptance:** an automaton accepts or rejects; it has no notion of "almost an identifier" or of a confidence, so a one-character error is a total miss.

### A.3 Estimated effect, reasoned from the Task 2 and Task 7 error rates (before measuring)

**Pattern recall.** Clean recall was 0.92. If an entity survives OCR with probability 1 − 0.35 = 0.65, recall on OCR text should be about 0.92 × 0.65 ≈ **0.60**; precision should change little, since a corrupted entity usually matches nothing rather than something wrong. **Analyser coverage.** Clean coverage was 98.6%. With 12% of ordinary words corrupted and about 5% of long words split, I expect OCR coverage around 0.986 × (1 − 0.12) ≈ **0.87** before counting the split words; for code-mixed text, if about one token in five is regional, coverage should fall to about **0.8**; for text with both, about 0.7. These estimates are compared with the measurements in B.

## B. Evaluation of the original design (2 marks)

### B.1 Three traces through the original pipeline

**OCR record** (first 260 characters)

| Clean text | Noisy text given to the pipeline |
|---|---|
| PAT523572: age 9 yrs 2C-B16. ICD M54.5. The cysts were ached near the biopsy. Felt and bled fast nephrology. The evidence reviews in the febrile clinician. Chronic bruises investigated in the allergy. Submitting bradycardia with anaemic strokes. The blood was | PAT523S72: age 9 yrs 2C-B16. ICD M54.5. The cysts were ached near the biopsy. Fe1t and bled fast nephrology. The evidence reviews in the febrile clinician. Chronic bruises investigated in the allergy. Submitting bradycardia with anaemic strokes. The blood was |

Entities the original patterns missed: patient_id "PAT523572"; ward_bed "2C-B16"; drug "Omeprazole 20 mg"; freq "three times daily"; bp "177/56". Spurious: icd "B16". Out-of-lexicon tokens: pat, fe, rder, o, three, times (6 of 88 word tokens).

**Code-mixed record** (first 260 characters)

| Clean text | Noisy text given to the pipeline |
|---|---|
| MRN-47333: age 27 years admitted on 18 Sep 2025 to Ward 7A Bed 4 with bacterium for 8 years. BP 125/86 mmHg, Temp 39.3 C. WBC 3.5 x10^9/L WBC=6.0gm%. ICD A09. Abnormal tenderness vomited on the severe lung. Injecting critical masks during chronic bandages. His | MRN-47333: age 27 years admitted nahi on thodaa 18 Sep 2025 to bohot Ward 7A Bed 4 with bacterium for 8 years. BP 125/86 mmHg, Temp 39.3 C. WBC 3.5 x10^9/L WBC=6.0gm%. ICD A09. Abnormal tenderness vomited durd on the severe lung. Injecting critical masks durin |

Entities the original patterns missed: admission_date "18 Sep 2025"; lab "WBC=6.0gm%"; discharge_date "21-04-2024"; phone "+91 62949 24421". Spurious: age "8 years"; duration "27 years". Out-of-lexicon tokens: nahi, thodaa, sep, bohot, durd, vali, aana, vaali, darad, irundhuchi, koduthen, ko (14 of 88 word tokens).

**OCR + code-mixed record** (first 260 characters)

| Clean text | Noisy text given to the pipeline |
|---|---|
| MRN-355165: age 28yo Ward 4B, bed 25. ICD N20.0. Mobilised and completed good tumours. No evidence of urologist, well stretcher or woman. The anaemic management fractures in the weak drain. Resolving afebrile doctors after tender drains. No evidence of renal w | MRN-355165: age ka 28y0 Ward 4B, bed 25. ICD N20.0. Mobilised ko and aana cornpleted good tumours kia. No evidence kya of dia urologist, well stretcher or woman kaicchal. The anaemic management fractures in ko the weak ka drain. Resolving afebrile dard doctors |

Entities the original patterns missed: age “28yo”. Spurious: phone “8705075633”. Out-of-lexicon tokens: ka, y, ko, aana, cornpleted, kia, kya, dia, kaicchal, ko, ka, dard (24 of 99 word tokens).

**What fails first, and why.** In the OCR record the first failure is the very first token: the patient ID “PAT523S72” does not match (S is not a digit), so the record cannot be linked to the patient; the next failure is “Fe1t”, an unknown word because the digit 1 ends the word token. In the code-mixed record the first failure is the regional particle “nahi”, which the analyser does not know, and every regional token behind it fails the same way. In the combined record the OCR error “cornpleted” (rn for m) fails at the word level while the regional tokens fail as before. Not every miss is caused by noise: textual dates (“18 Sep 2025”), spaced phone numbers and “28yo” are also missed on the clean text (Task 2) and are format-variation errors of the original patterns.

### B.2 Measured degradation against the clean baseline (same 300 records)

| Sample | Text | Pattern precision | Pattern recall | Analyser coverage | Retrieval F1 (raw) | Retrieval F1 (lemma) |
|---|---|---|---|---|---|---|
| OCR only | clean | 0.90 | 0.93 | 98.9% | 0.79 | 0.99 |
| | noisy, original | 0.87 | 0.63 | 80.3% | 0.73 | 0.94 |
| Code-mixed only | clean | 0.88 | 0.91 | 98.4% | 0.82 | 0.99 |
| | noisy, original | 0.88 | 0.88 | 69.3% | 0.82 | 0.99 |
| Both | clean | 0.91 | 0.91 | 98.6% | 0.83 | 0.99 |
| | noisy, original | 0.87 | 0.61 | 57.5% | 0.77 | 0.94 |
| All 300 | clean | 0.90 | 0.92 | 98.6% | 0.81 | 0.99 |
| | noisy, original | 0.87 | 0.71 | 67.8% | 0.77 | 0.96 |

Over all 300 records, pattern recall fell from 0.92 to 0.71 (estimate before measuring: ≈ 0.60 for OCR-only text, measured 0.63), precision barely changed (0.90 to 0.87), analyser coverage fell from 98.6% to 67.8% (estimates 0.87, 0.8 and 0.7 for the three groups; measured 80.3%, 69.3%, 57.5%), and lemma-search F1 fell from 0.99 to 0.96. The estimates were close for patterns and for the combined group, and optimistic for OCR coverage (hyphen splits create two unknown tokens each).

A notable result: code-mixing alone leaves retrieval F1 unchanged (0.99 → 0.99), because the English clinical words that carry the search terms are still analysed, even though coverage falls to 69%; **coverage is not a good predictor of retrieval quality on its own.** OCR noise, in contrast, destroys search terms and entities.

**B.3 Would a different formalism be more economical?**

| Formalism | What it would solve | Economy compared with extending the FST |
|---|---|---|
| Edit-distance matching against the lexicon (weighted: OCR confusions cost less) | Misspelt and OCR-corrupted English words | Much cheaper: needs only the existing lexicon and a small confusion table; extending the FST with every possible corruption would multiply its size. |
| Weighted automata / weighted transducers | Replacing exact accept/reject by costs; the principled form of the same edit-distance idea | Best fit in theory; I implemented the same idea through candidate generation to stay within the standard library. |
| Character n-gram language identifier | Deciding whether an unknown token is regional or English | Cheap, needs only short word lists; much cheaper than writing Tamil or Hindi morphology. |
| Extending the FST with regional morphology | Properly analysing regional words | Expensive and needs linguists; not justified when search needs only the English clinical terms. |

## C. Adaptation or redesign (2 marks)

**Decision: MODIFY.** The finite-state core (patterns, DFAs, lexicon FST, rule cascade, Porter fallback) is kept, because its behaviour on clean English is good and the new problems are at its edges. A front end is added that repairs the input, and a language-identification step is added before analysis.

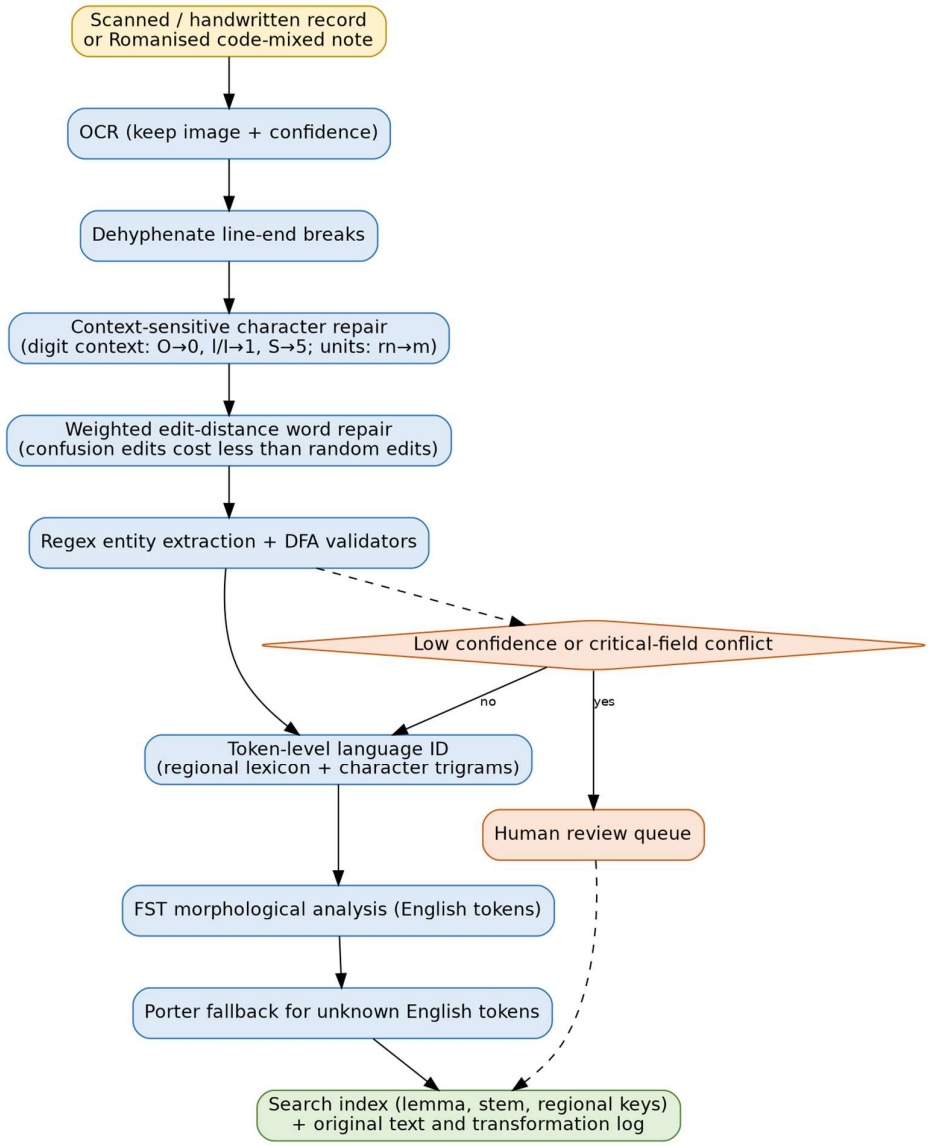



*Figure 10. Revised design. Original text is always retained; low-confidence or critical-field conflicts go to human review.*

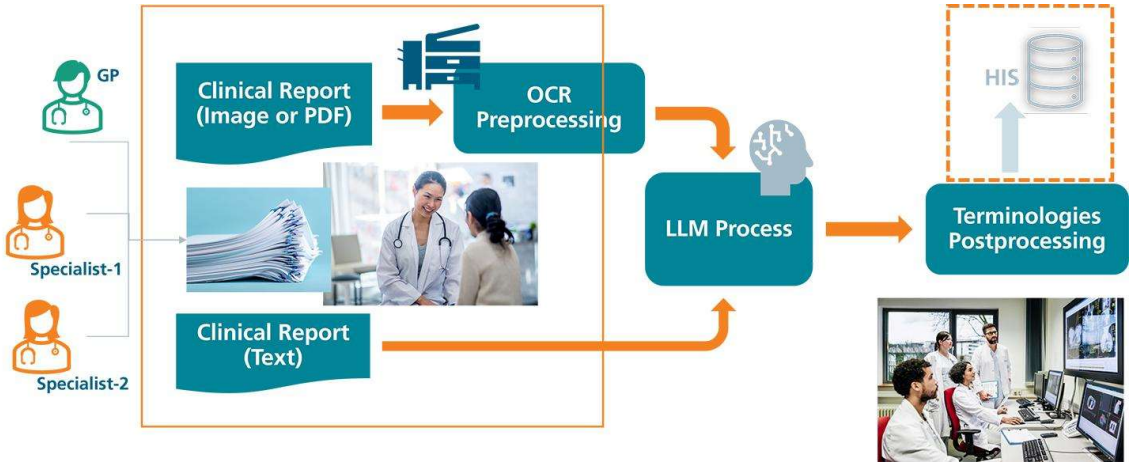



*Figure 11. Reference architecture supplied with the assignment materials for ingesting scanned and text reports (OCR preprocessing → LLM process → terminology post-processing → hospital information system). The finite-state front end of Figure 10 would sit in the "OCR preprocessing" and "terminology post-processing" boxes.*

### C.1 Exactly what changes

| Component | Change |
|---|---|
| Patterns | **Not changed.** Repair is done before matching, so the strict patterns still accept only valid shapes; widening them to accept O and l in digit positions would raise false positives on ordinary words. |
| Automata | Unchanged; they now see repaired text. Their accept/reject result is used as a **validator**: a repaired string that still fails its DFA (for example an ID with five digits) is sent to human review. |
| Lexicon | Added a regional lexicon of 25 lexemes (50 known spellings) and unknown-English handling; the English lexicon is unchanged. |
| Rule cascade | Unchanged. A dehyphenation rule (letter-hyphen-newline-letter → joined) and context-sensitive character rules were added **before** it: in numeric context O,o→0, I,l→1, S→5, B→8; before a unit, rn→m, I→l (so "5OO rng" → "500 mg"). |
| New: word repair | Single-edit candidate generation: confusion substitutions first, then one insertion, deletion, substitution or transposition; accepted only if exactly one candidate is a known word (a cheap form of weighted edit distance). |
| New: language ID | A token not in the lexicon is labelled regional if it is a known regional word or if a character-trigram naive Bayes model (trained on the English lexicon and the known regional spellings) prefers the regional class. |
| New: fallback | Unknown English tokens receive a Porter stem key; regional tokens are indexed as themselves (variant spellings are normalised through the lexicon). |

### C.2 Measured effect

| Sample | Design | Pattern precision | Pattern recall | English analysis coverage* | Retrieval F1 |
|---|---|---|---|---|---|
| OCR only | original | 0.87 | 0.63 | 80.3% | 0.94 |
| | redesigned | 0.90 | 0.78 | 95.2% | 0.99 |
| Code-mixed only | original | 0.88 | 0.88 | 69.3% | 0.99 |
| | redesigned | 0.88 | 0.87 | 69.4% | 0.99 |
| Both | original | 0.87 | 0.61 | 57.5% | 0.94 |
| | redesigned | 0.90 | 0.75 | 67.0% | 0.99 |
| All 300 | original | 0.87 | 0.71 | 67.8% | 0.96 |
| | redesigned | 0.89 | 0.80 | 75.3% | 0.99 |

**Share of word tokens analysed by the English analyser, directly or after repair; regional tokens, correctly identified and indexed as themselves, are counted as handled but not as English-analysed, so for the code-mixed groups this figure stays near 0.7 by design.*

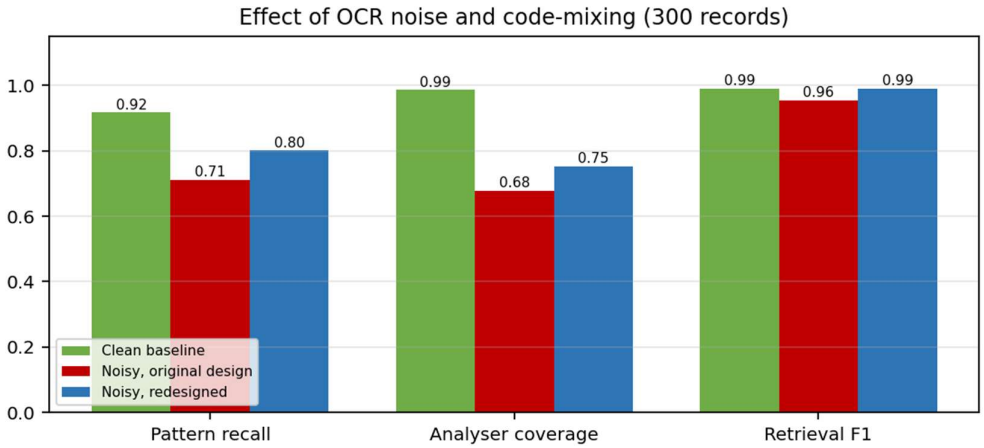



*Figure 12. Clean baseline, noisy input through the original design, and noisy input through the redesign.*

| Item | Result |
|---|---|
| Language identification | 98.9% of 4,013 regional tokens recognised; 95.9% of 1,073 tokens in the **held-out spellings** recognised by the character model alone; 0.0% of 4,667 English words wrongly labelled regional. |
| Run time | 0.50 ms per record (original: extract + lemma index) against 1.83 ms per record (redesign), about 3.6 times slower, still milliseconds per record. |
| Annotation effort to maintain | Regional lexicon: 25 lexemes, about 8–10 h with a speaker (estimate); confusion table: 2 h; a continuing review of the human-review queue. The regional list must grow with the hospital's languages. |

## D. Evidence-based defence (2 marks)

**Why MODIFY and not RETAIN or REDESIGN, from the evidence:**

- **Not retain:** under the original design, pattern recall fell to 0.71 (OCR-only: 0.63) and retrieval F1 on OCR text to 0.94; patient identifiers, the field most critical for safety, were lost whenever a confusion fell on them (35% of ID strings by construction of the simulation).
- **Not redesign:** the modified design recovers OCR-only retrieval F1 to 0.99 and pattern recall from 0.63 to 0.78 using the existing lexicon and automata; the clean-text behaviour is unchanged. A full replacement by a learned model would need thousands of labelled OCR/code-mixed examples, which the hospital does not have yet (Task 11).
- **The regional layer is cheap and effective for search:** retrieval stays at 0.99 for code-mixed text because search terms are English; language identification recognises 98.9% of regional tokens, 95.9% even for spellings it has never seen.
- The automata keep their role as strict validators; their exact accept/reject behaviour is what lets the system detect that a repaired field is still wrong and route it to a human, instead of guessing.

**Remaining limitations, including what a finite-state layer will never handle however far it is extended:**

- Pattern recall after repair is 0.80, still below the clean 0.92: some corruptions are not repairable (letters in IDs, merged characters, digits changed to other digits such as 3 ↔ 8), and a wrong digit that looks valid is an **undetectable dosage error**. Critical numeric fields must therefore keep a human check; no confidence threshold is perfect.

- Corrected words are accepted only when exactly one candidate exists; ambiguous corruptions are left alone and fall to the Porter fallback.
- **Unbounded structure:** nesting and long-distance agreement are beyond any finite-state machine (Task 4.5).
- **Meaning and context:** negation scope ("no evidence of infection"), the sense of "OD" or "discharge", and the difference between an age and a duration need syntax, semantics and discourse, not morphology.
- **Open vocabulary:** unknown drugs, eponyms and regional morphology (Tamil and Hindi are agglutinative or rich in inflection) cannot be listed in advance; a learned model with subword units is the realistic way to extend coverage.
- This evaluation uses simulated OCR and code-mixing; before deployment it must be repeated on a de-identified sample of real scans and real clinician notes.

## Conclusion

A finite-state layer of regular expressions, automata, a morphological transducer with a spelling-rule cascade, and a Porter fallback solves the problem the hospital described: on the synthetic corpus it lifts retrieval F1 from 0.81 (exact match) to 0.89 with Porter stemming and to a ceiling of 0.99 with full analysis, while preserving the original text and keeping derivational families apart. The Day-4 experiment shows where the formalism strains: simulated OCR noise reduced pattern recall from 0.92 to 0.71, and a front end of repair and language identification restored retrieval to 0.99 at a cost of a few milliseconds per record. The recommended design is therefore hybrid, auditable and supervised by human review for critical fields, with the explicit expectation that real-data figures will be lower than the ceiling reported here.